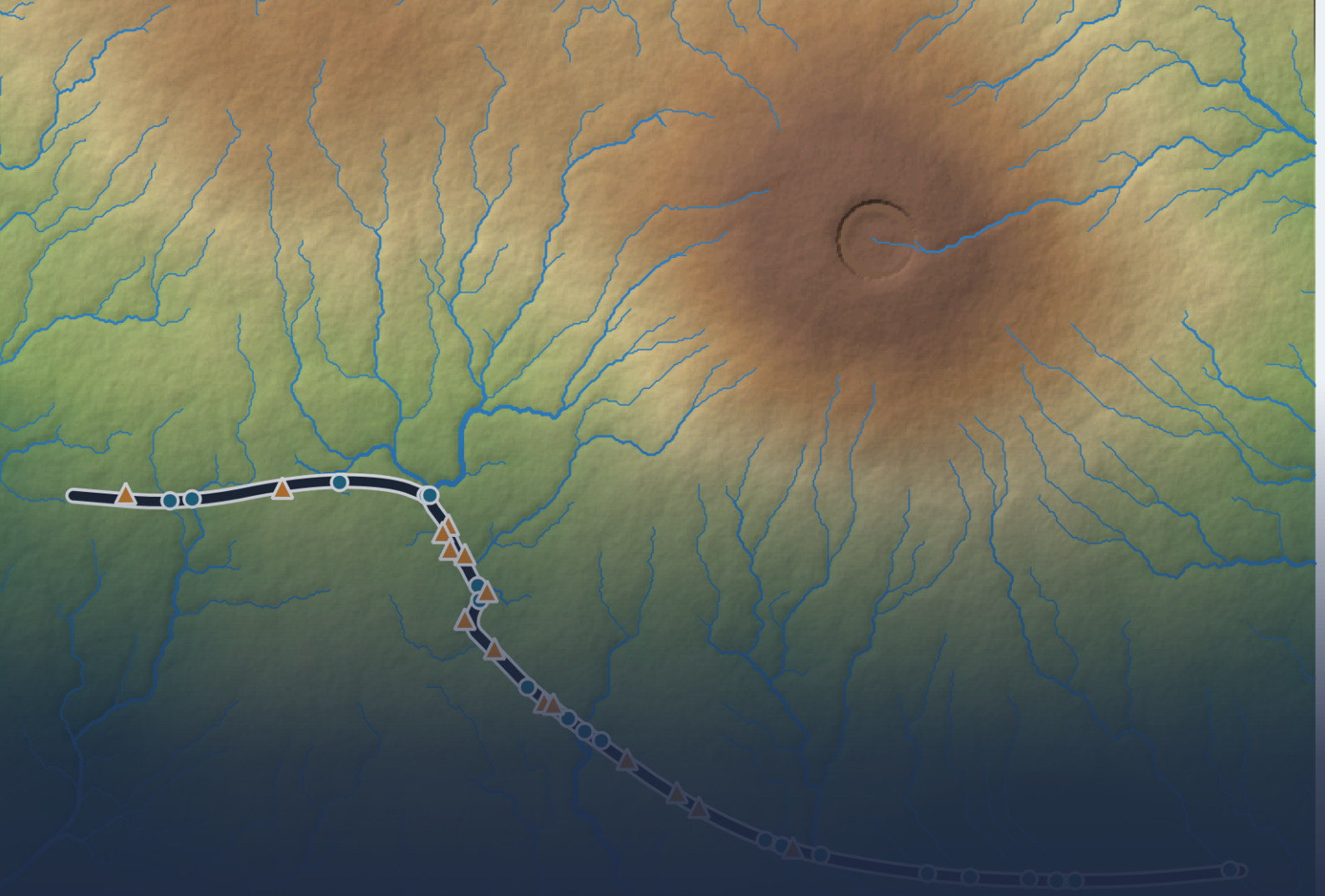


QGIS PLUGIN · VERSION 1.0.1

QEHT — QGIS Engineering Hydrology Toolkit

User Guide and Technical Manual

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QGIS	3.34 to 4.x (Qt5 and Qt6 builds)
Licence	GNU General Public License v2 or later
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Every figure in this guide was produced by the plugin from a synthetic demonstration site: invented terrain, road, soils, land cover, rainfall and river names at an invented location. No project data is shown.

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0 What QEHT does

QEHT turns a digital elevation model (DEM) into the drainage quantities a road or catchment study needs: conditioned terrain, flow direction and accumulation, streams, catchments, longest flow paths and their slopes, crossings along a road, and the soils, runoff, rainfall, erosion and time-of-concentration inputs of a design-flood calculation. Everything runs inside QGIS, using only NumPy and the GDAL bindings that QGIS already ships. It starts no external program and needs no extra Python package, so it works on managed workstations where tools that start external programs are blocked.

Terrain and drainage

DEM preparation and checks, priority-flood fill, D8 with Barnes (2014) flat resolution, accumulation, Strahler streams, catchments, longest flow paths and four named slopes.

Road drainage

Crossing candidates with chainage, embankment burning, the ground profile, a coverage check that proposes missing crossings, channel sections, floodplain width, mapped-river check.

Design inputs

Soil texture and USLE K from any soil source, curve numbers and Rational C, rainfall zones, five time-of-concentration methods side by side, land-cover scenarios.

Erosion

SPI, TWI, LS and RUSLE with source-tagged factors, severity classes, erosion along the road and a side-drain siltation indicator.

Uncertainty and QA

Flat-method sensitivity per crossing, DEM-error Monte Carlo ranges of area and Tc, undeclared-NoData and resampling audits, provenance in every output.

Delivery

One-click pipeline into one folder: layers, tables, an HTML run report, KMZ for Google Earth, an XLSX workbook and a self-describing GeoPackage.

Rules the plugin keeps. No engineering value ships as a default when it is not known: a land-cover proxy is labelled PROXY, a relation used outside its origin is labelled ESTIMATE, and both labels travel into every table, report and package. Where a design method needs a choice (which slope, which Tc method, which flat method), QEHT reports the alternatives and records which was used; it does not choose for you.

PART A

Getting started

Install the plugin, prepare a DEM it can trust and the optional inputs, and meet the demonstration site used throughout this guide.

1 Overview

QEHT is a QGIS Processing provider. Its tools appear in the Processing Toolbox under **Engineering Hydrology** and behave like any other Processing algorithm: they validate their inputs, run in batch mode, chain in the Graphical Modeler, log to the history and can be called from Python with `processing.run()`.

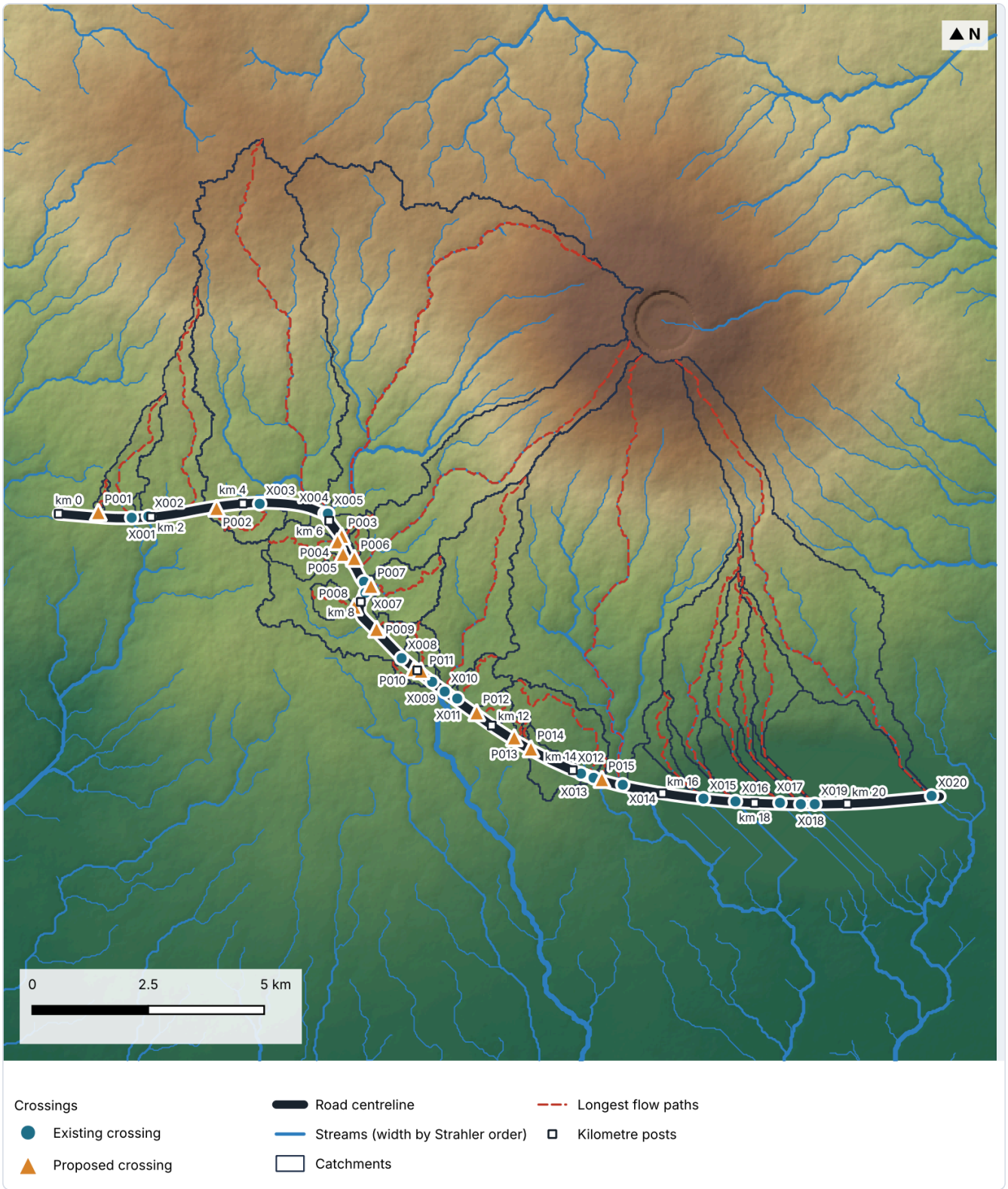


Figure 1. The demonstration site after one pipeline run: 22 km of road with 20 existing crossings (blue) and 15 proposed ones (orange), their catchments and longest flow paths, and streams of at least 0.5 km² drawn by Strahler order over the relief quicklook. The road crosses the slopes of a volcano, runs 2.5 km beside a main valley and ends on the flat bed of an old lake.

Toolbox group	Tools
Terrain and drainage	Prepare DEM for hydrology · Fill depressions · D8 flow direction · Flow accumulation · Stream network and Strahler order · Stream network to polylines · Delineate catchments from pour points · Longest flow path · Catchment and flow path characteristics

Toolbox group	Tools
Road drainage	Road crossing candidates · Burn crossings through embankments · Alignment ground profile · Drainage coverage check along a road · Clip DEM to the road's contributing area · DEM uncertainty at crossings
Soils and erosion	Soil, rainfall and runoff parameters for catchments · Erosion indices and RUSLE soil loss · Sample erosion along alignment
Interoperability	Build design hydrology package · Renumber and relink exchange package
Workflow	Run hydrology pipeline (one click) · Run report from a design hydrology package · Export package to KMZ / XLSX

A typical workflow

1. **Prepare the DEM**: one product, projected in metres, resampled bilinearly, NoData declared (section 3 and 10).
2. **Run the pipeline** with the DEM and the road centreline, and stop after the crossing candidates (section 5).
3. **Review the candidates**: accept, reject, move or add crossings; then run again with the reviewed layer (section 6).
4. **Read the run report**: DEM checks, flat-sensitive crossings, coverage findings, uncertainty ranges and warnings (section 8).
5. **Adopt or delete proposed crossings** and relink the package if you edited it (section 29).
6. **Take the numbers into design**: the characteristics table, the Tc table and the XLSX workbook, with the slope and Tc method you use stated in the design report.

2 Installing and updating

From the QGIS plugin repository

1. Open **Plugins → Manage and Install Plugins...**
2. Search for **QEHT** and press **Install** (or **Upgrade**).
3. Open **Processing → Toolbox** and expand **Engineering Hydrology**.

From a ZIP file

Use **Plugins → Manage and Install Plugins → Install from ZIP** with `qeht-1.0.1.zip`. The zip holds one folder, `qeht`; keep that name, because the plugin imports `qeht.core`.

> **REQUIREMENTS**
> QGIS 3.34 or later, including QGIS 4 (Qt6), on Windows, macOS or Linux. Nothing else: no SAGA, GRASS, WhiteboxTools, TauDEM or pip packages. Matplotlib, which QGIS ships on most installs, is used only for the KMZ legend and the optional profile charts.

Memory

QEHT holds the whole grid in memory. Every tool estimates its peak before it runs and warns when the estimate approaches the free RAM. As a guide, the routing steps need about 50–170 bytes per cell: a 4,000 × 4,000 DEM (16 million cells) needs 1–3 GB. For a national mosaic, clip to the road's contributing area first (section 20).

3 Preparing the inputs

The DEM

Every result depends on the DEM, so most of the checks in QEHT are about it. A DEM QEHT can trust is:

- **Projected, in metres.** A geographic DEM (degrees) is refused before any computation. **Prepare DEM for hydrology** reprojects to the local UTM zone (section 10).
- **One product at its native resolution.** FABDEM, Copernicus GLO-30 and ALOS AW3D30 are 30 m. Mixing products or exporting finer than native adds no information.
- **Resampled bilinearly, never nearest neighbour.** Nearest neighbour repeats whole rows and columns wherever the output grid is finer than the source. The repeats create exact ties and artificial flats, and routing across them is decided by the flat method rather than the terrain.
- **With its NoData declared.** A block of zeros with no NoData in the header is read as a sea at 0 m that the whole catchment drains into.

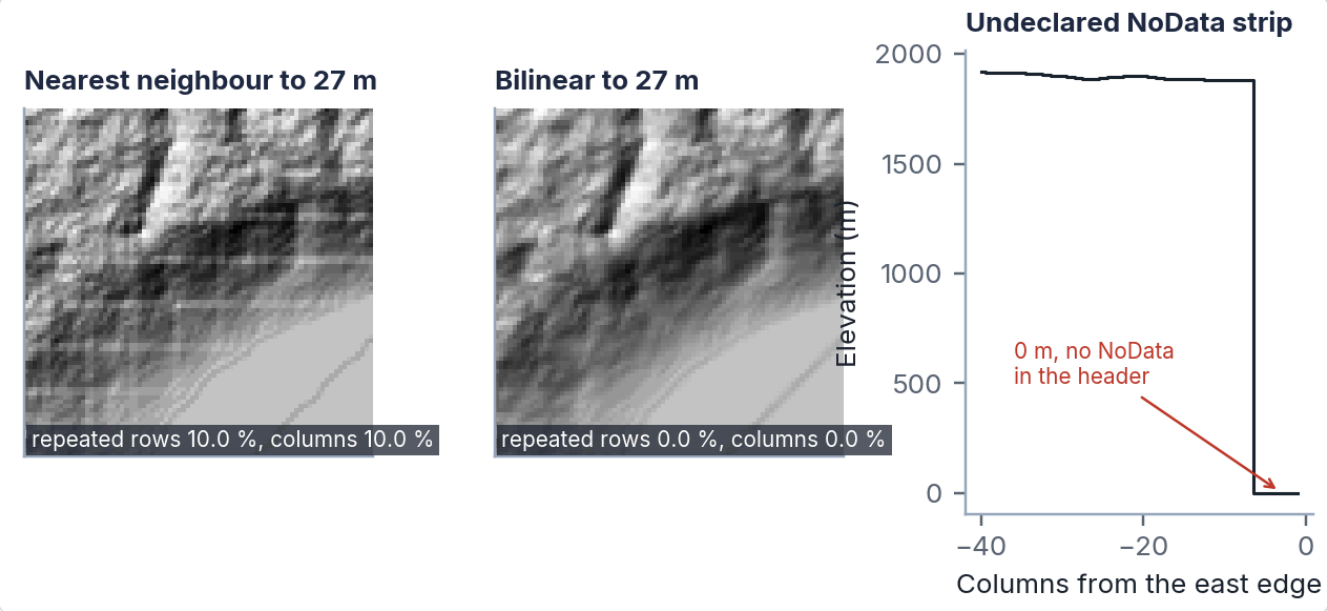


Figure 2. The two DEM defects QEHT audits. Left and centre: the demonstration DEM regridded from 30 m to 27 m. Nearest neighbour repeats one row and column in ten (the stripes in the hillshade); bilinear repeats none. Right: the last 40 columns of one row of the demonstration DEM, which ends in a strip of zeros that the file does not declare as NoData. Fill, D8 and the pipeline print a warning for both, with the share of cells affected.

When the audit reports undeclared NoData, give the value in **Treat this DEM value as NoData** (pipeline) or **Treat this value as NoData** (Fill depressions). When it reports repeated rows or columns in the *input*, the damage happened before QEHT: get the native product and prepare it again. A FABDEM export from Earth Engine at a scale slightly finer than its native 1″ grid (for example 0.000269°) repeats about one row and column in 33; export in the native projection and scale.

The road centreline

A line layer, in any CRS (it is transformed to the DEM's). Chainage runs along the parts in layer order and the digitising direction from the **Start chainage**; tick **Reverse chainage direction** to run it the other way. A rough digitised centreline is enough before the geometric design exists: crossings, chainages and the ground profile are then provisional and are re-run on the design alignment later.

Optional inputs

Input	Global source (national data is better)	Used for
Soil map or rasters	SoilGrids 2.0 (ISRIC, 250 m); HWSD v2 units; SOTWIS / SOTER databases	Texture, USLE K, hydrologic soil group proxy (section 22)
Hydrologic soil group	HYSOGs250m (ORNL DAAC)	Overrides the texture-based group
Land cover	ESA WorldCover 2021 (10 m)	Curve number, Rational C, C factor, Tc retardance (sections 23, 25, 26)
Rainfall zones	A national design-rainfall zoning	Zone per catchment (section 24)

Input	Global source (national data is better)	Used for
Mean annual rainfall	CHIRPS climatology	MAP per catchment; R estimate
Rainfall erosivity R	GloREDa (ESDAC)	RUSLE (section 26)
2-yr 24-h rainfall P2	A national IDF study	TR-55 sheet-flow Tc (section 25)
Mapped waterways	OpenStreetMap waterways, HydroRIVERS	Check against mapped drainage (section 19)

4 The demonstration site

Every figure in this guide comes from one pipeline run on a synthetic site that ships with the repository. Nothing in it describes a real place. `tools/dev/make_example_dem.py` builds the DEM (`examples/example_dem.tif`): 760 × 720 cells at 30 m in EPSG:32737 at an invented origin, with a volcanic cone and crater, dendritic valleys carved by QEHT's own flow accumulation, a low plain stored in whole metres like ALOS, and an undeclared NoData strip along the east edge. `tools/guide/make_demo_site.py` adds the rest:

File	Content
`demo_dem.tif`	The example DEM with an old lake bed on the south-east plain: a plane falling 0.15 % in whole metres, so it has real flats.
`demo_road.gpkg`	"Demo Road", 22 km: across the volcano's slopes, 2.5 km beside a main valley, then over the lake bed.
`demo_landcover.tif` , `demo_landcover_2040.tif`	WorldCover classes from elevation, slope and noise, with a town on the road; the 2040 scenario turns cropland within 1 km of the road into built-up land.
`demo_soils.gpkg`	Three soil units with `sand` , `silt` , `clay` , `oc` , `drain` and `hsg` fields.
`demo_rain_zones.gpkg` , `demo_map.tif` , `demo_r.tif`	Two rainfall zones, mean annual rainfall 760–1,140 mm/yr, and an erosivity raster.
`demo_rivers.gpkg`	"Mapped" rivers of at least 8 km², traced on the DEM plus a smooth error field so that they depart from the DEM streams the way a real map does.

```
python3 qeht/tools/guide/make_demo_site.py demo_site
```

Run it from the folder that contains the `qeht` repository with the Python that QGIS uses (on Windows, the OSGeo4W shell). The pipeline run behind the figures used the parameters in Figure 3 and took under two minutes on a laptop, including 30 DEM-uncertainty realisations.

PART B

The one-click pipeline

One tool runs the whole chain, from DEM checks to the run report, and writes one folder. Each step is the stand-alone tool of the same name, so the results are the same as running the tools one by one.

5 Running the pipeline

Open **Run hydrology pipeline (one click)** in the Workflow group. Only the DEM and an output folder are required; with a road the pipeline also finds the crossings, and every optional input adds a block of results.

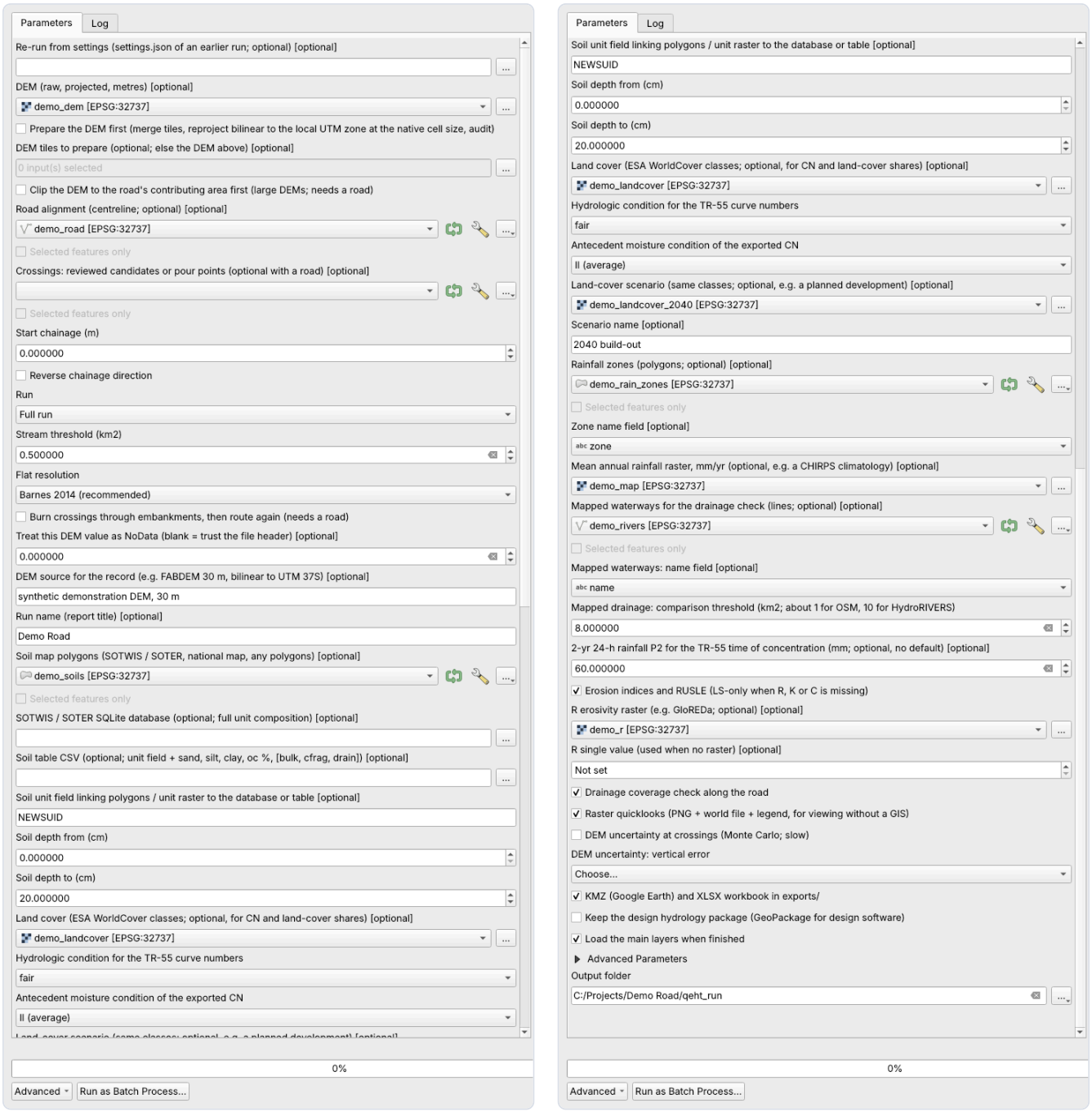


Figure 3. The pipeline dialog filled in for the demonstration run, at two scroll positions. In QGIS a help panel on the right lists the steps and the output layout.

Parameter	Default	Guidance
DEM (raw, projected, metres)	—	The raw DEM. QEHT conditions it itself; reported elevations and slopes always come from the raw DEM.
Prepare the DEM first	off	Merges tiles from *DEM tiles to prepare* (or uses the DEM), reprojects bilinearly to the local UTM zone and audits (section 10).
Clip the DEM to the road's contributing area first	off	For DEMs too large for memory (section 20). Needs a road.
Road alignment, Start chainage, Reverse	0 m	With a road, crossings are found where streams cross it.
Crossings	—	Reviewed candidates (section 6) or your own pour points. Pour points are snapped to the nearest stream within 5 cells.

Parameter	Default	Guidance
Run	Full run	*Stop after crossing candidates* writes the candidates for review and stops.
Stream threshold (km²)	1.0	The smallest area that is a stream. Set it to the smallest catchment you would design a crossing for; 0.5 km² was used here.
Flat resolution	Barnes 2014	Keep Barnes unless you have a reason; section 11 explains the choice.
Burn crossings through embankments	off	For DSMs and survey DEMs that show the road embankment as a dam (section 15).
Treat this DEM value as NoData	blank	Set it when the DEM checks report undeclared NoData (0 for the demonstration DEM).
DEM source, Run name	—	Recorded in the metadata and the report title.
Soil, land cover, scenario, rainfall inputs	—	Part E. Each block is computed only when its input is given.
Mapped waterways, name field, comparison threshold	1 km²	Section 19. About 1 km² for OpenStreetMap, 10 km² for HydroRIVERS.
2-yr 24-h rainfall P2	none	Only TR-55 sheet flow needs it. Without it TR-55 Tc is left empty.
Erosion indices and RUSLE	on	Without R, K or C the erosion step gives terrain indices and LS-only classes.
Drainage coverage check	on	Proposes crossings for sags and uncovered streams (section 17).
DEM uncertainty at crossings	off	Monte Carlo; needs a vertical error preset or value (section 21).
Quicklooks, KMZ and XLSX	on	Viewing without a GIS (sections 7 and 9).
Keep the design hydrology package	off	The self-describing GeoPackage of Part F. The stand-alone outputs are the same either way.

What runs, in order

Prepare DEM (optional) → clip to the contributing area (optional) → DEM checks → fill → D8 → accumulation → streams → crossing candidates (with a road) → burn and route again (optional) → erosion indices and RUSLE → crossings, catchments and flow paths with the soil, runoff, rainfall, scenario and erosion blocks → coverage check and proposed crossings → floodplain width → channel sections → mapped-drainage check → DEM uncertainty (optional) → time of concentration → side-drain siltation → quicklooks → flat-method check → tables, report and exports.

The log lists every step with its warnings. Read it once: warnings about the DEM, flat-sensitive crossings and inputs outside a method's range are repeated in the run report, but the log also shows the memory estimate and the time per step.

6 Reviewing crossings and running again

QEHT finds where streams cross the road. It does not know where the culverts are, which streams the road actually blocks, or which crossings a designer will combine. Review the candidates before the full run:

1. Run with **Run = Stop after crossing candidates**. The folder holds `layers/crossing_candidates.gpkg` and `settings.json`.
2. Open the candidates over imagery or survey. Set `status` to `accepted` or `rejected`; move a point onto the culvert; add points where there is a structure QEHT missed. Candidates you leave as `candidate` are ignored when any is accepted; when none is accepted, the recommended ones (`recommended = 1`) are used.
3. Run again with **Re-run from settings** = the first run's `settings.json`, **Crossings** = the edited layer and **Run = Full run**. Every other parameter comes from the settings file; a new output folder overrides the old one.

A candidate you move by more than two cells is snapped from its new position, and its chainage is taken from the road, exactly like a point you add. A candidate you leave in place keeps the outlet cell QEHT found for it.

YOUR OWN POUR POINTS

Any point layer works as *Crossings*: surveyed culvert positions, a structure inventory, points digitised from imagery. The pipeline numbers them X001, X002 … along the chainage (downstream first without a road). To keep your own numbering as `outlet_uid`, run **Build design hydrology package** with *ID attribute* set (Part F). Points are snapped to the nearest stream cell within 5 cells; check `snap_dist_m` in the crossings layer for points that moved far.

7 The output folder

Folder	Contents
`layers/`	crossings, catchments, flow paths, streams (reaches with Strahler order), crossing candidates, parallel reaches, coverage findings, sag points, flat stretches, the alignment profile and the road (one GeoPackage each)
`rasters/`	filled DEM, flow direction (D8 codes), flow accumulation, streams, Strahler order, `erosion/` (indices, RUSLE, classes and `erosion_run.json`); `dem_input.tif` when a NoData value was declared
`tables/`	`catchment_characteristics.csv` (one row per crossing), `time_of_concentration.csv` and one CSV per layer, including the run metadata and the field dictionary
`quicklooks/`	PNG, world file and legend JSON of the relief, flow accumulation and erosion rasters, at most 4,096 px
`report/`	`run_report.html`
`exports/`	`design_hydrology.kmz` and `design_hydrology.xlsx`
`package/`	`design_hydrology.gpkg`, when *Keep the design hydrology package* is ticked
`settings.json`	every parameter of the run, for an exact re-run

The characteristics table is the hand-off to design. Its columns come in groups (identification and location, area, the four slopes, flow-path segments, channel slopes, Tc, channel section, flat check, soils, curve number, rainfall, shape, erosion, uncertainty), and a group appears only when at least one crossing has a value. Rows are ordered by chainage.

8 The run report

`report/run_report.html` opens in any browser and prints to PDF for an appendix. It reads the package only; nothing is recalculated, so the report always matches the data.

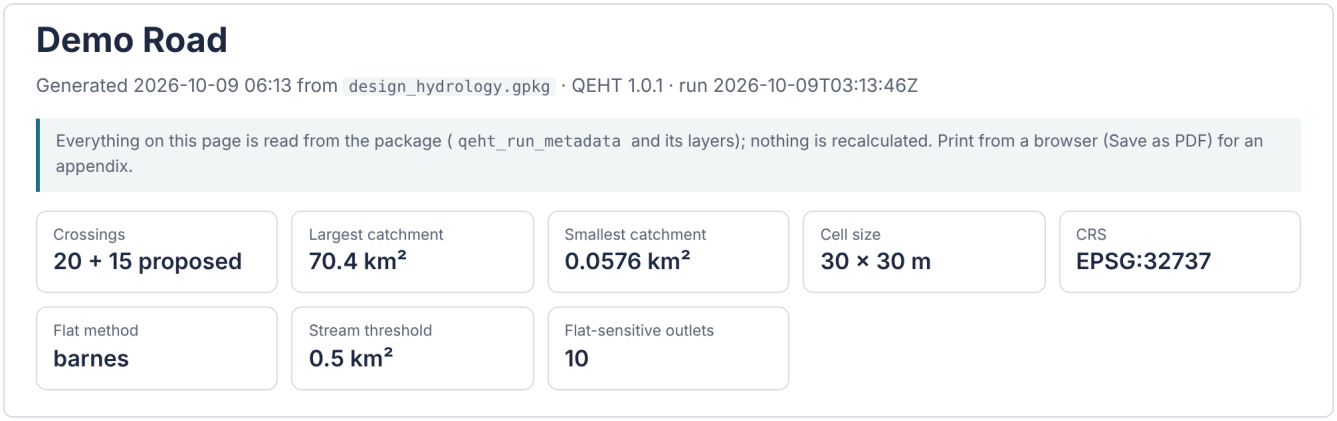
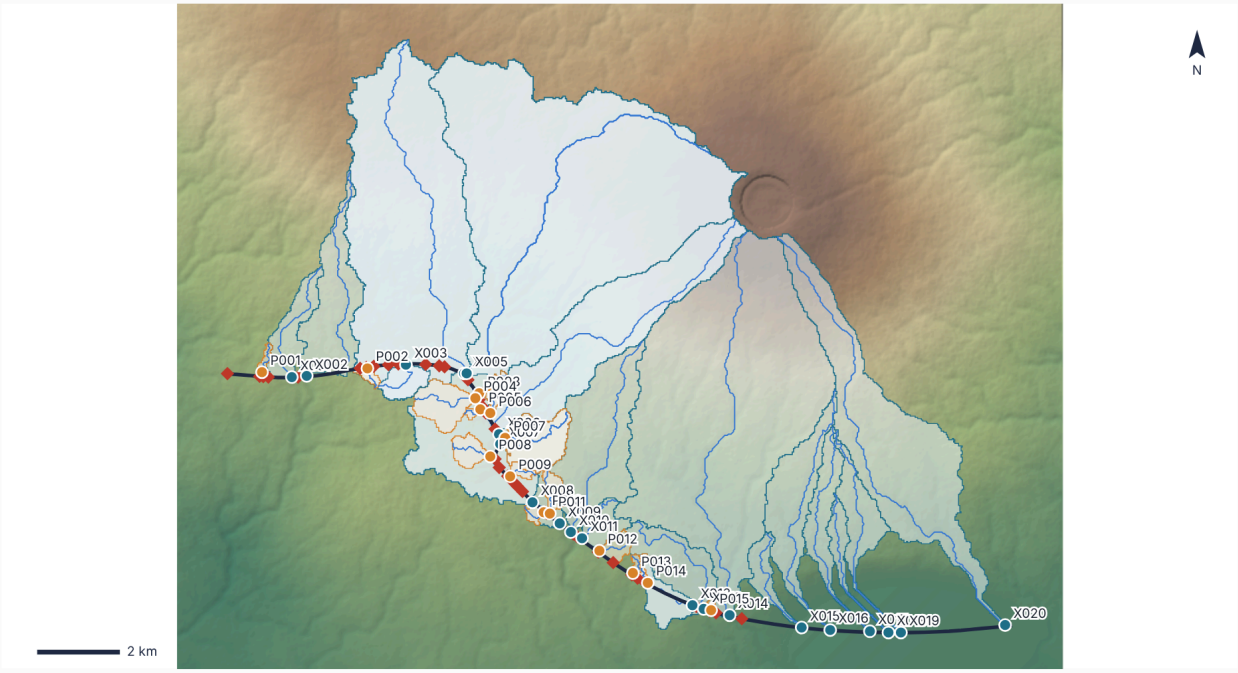


Figure 4. The report header: crossings, the largest and smallest catchments, grid, CRS, flat method, stream threshold and the number of flat-sensitive outlets.

Plan



Schematic plan from the package geometry: catchments (blue; proposed orange), longest flow paths, road and crossings over the terrain (hillshade of the DEM). Drawn to scale.

Figure 5. The plan, drawn to scale from the package geometry over the relief quicklook: catchments (proposed ones outlined in orange), flow paths, the road, crossings and coverage findings.

The sections that follow are, in order:

- **Crossing schedule**: one row per crossing with chainage, area, flow-path length, 10–85 slope, catchment slope, curve number, rain zone, rainfall and the flat check.
- **DEM checks**: the NoData and resampling audits, the flat-method check (section 11) and channel sections of low quality.
- **DEM uncertainty**, **Land-cover scenario**, **Check against mapped drainage**, **Time of concentration** and **Drainage coverage along the road**, when those blocks ran.
- **Inputs and provenance**: the DEM path and SHA-256, the QEHT version, every dataset and lookup, and the labels of proxies and estimates.
- **Warnings**, the **curve number lookup used** and the list of output files.

DEM checks

Resampling audit: no repeated rows or columns, no NoData problems.

Flat-method check

Contributing area under both flat methods (largest within one cell of the outlet). A flagged crossing's area depends on how flats are routed: verify it against mapped drainage or on site.

10 of 35 crossing(s) flagged (the others agree within the tolerance; values in the characteristics table).

outlet_uid	area_barnes_km2	area_toward_km2	flat_sensitivity_pct	flag
P003	0.0594	52.929	99.888	verify
P005	54.236	0.0639	99.882	verify
P009	0.139	66.781	99.792	verify
P006	0.249	54.485	99.542	verify
X006	54.445	0.312	99.426	verify
P004	52.919	1.184	97.762	verify
P007	1.696	65.613	97.416	verify
P001	0.124	0.054	56.522	verify
X004	31.711	52.547	39.652	verify
X005	31.711	52.547	39.652	verify

Figure 6. DEM checks and the flat-method check. Ten crossings change area by more than 10 % between the two flat methods: those on the lake bed and in the valley beside the road. Their catchments are decided by flats rather than terrain and need checking on site or against a map.

9 KMZ and XLSX exports

Export package to KMZ / XLSX (Workflow), and the pipeline by default, write two files for people without a GIS. Both writers are pure Python.

KMZ for Google Earth. Coordinates are transformed to WGS 84. Existing crossings are blue and proposed ones orange; each has a card with its location and chainage, catchment area, slopes, elevations, soil, curve number, rain zone, Tc by every method with its validity, the channel section and the erosion values. Catchments are translucent polygons, the road carries kilometre posts, and coverage findings, channel sections, side-drain siltation reaches and the relief quicklook are separate folders that can be switched on.

XLSX workbook. Five sheets: *Catchment characteristics*, *Time of concentration*, *Coverage findings*, *Run metadata* and *Field dictionary*. The characteristics sheet runs in design order (identification, area, flow path, elevations, the four slopes, segments, channel slopes, Tc, CN and C, rainfall, soil, widths, flags), with units in row 2 and panes frozen at B3. Every column is always written, so the layout is fixed; the column order is one list in `core/report/xlsx_layout.py` and can be matched to a house workbook.

PART C

Terrain and drainage

The stand-alone tools behind the pipeline's first steps. Use them to run a step on its own, to compare settings, or to build your own chain in the Graphical Modeler.

10 Prepare DEM for hydrology

Most bad runs start before QEHT: a DEM reprojected with nearest neighbour, an export finer than the native grid, or NoData zeros the file does not declare. **Prepare DEM for hydrology** does the preparation in-process.

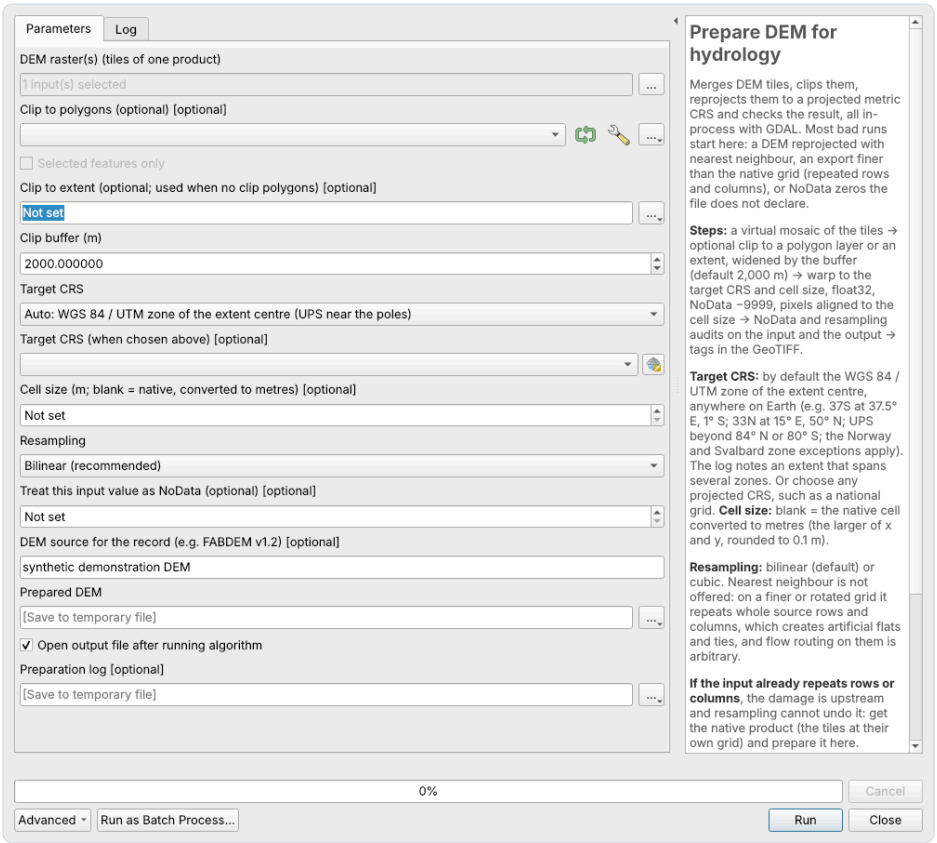


Figure 7. Prepare DEM for hydrology.

1. **Mosaic.** The tiles in *DEM raster(s)* become one virtual mosaic. They must be one product in one CRS.
2. **Clip.** To *Clip to polygons* or *Clip to extent*, widened by the *Clip buffer* (2,000 m), so catchments that start outside the area of interest are not cut.
3. **Reproject.** *Target CRS* = *Auto* picks the WGS 84 / UTM zone of the extent centre anywhere on Earth (UPS beyond 84° N and 80° S; the Norway and Svalbard zone exceptions apply). Or choose any projected CRS, such as a national grid. *Resampling* is bilinear (recommended) or cubic; nearest neighbour is not offered. The *Cell size* defaults to the native cell converted to metres.

4. **Audit.** The input and the output are checked for undeclared NoData and repeated rows and columns. Repeats in the input mean the damage is upstream.
5. **Record.** The output is float32 with NoData −9999. The DEM source and the preparation record are written into the GeoTIFF tags, and the package reads them into its metadata. The optional JSON log records the inputs, CRS, cell size, resampling, both audits and an estimate of the native grid.

11 Fill, flow direction and flats

Fill depressions

Closed depressions are filled by priority flood (Barnes, Lehman and Mulla 2014). Most are DEM noise; some are real (a crater, a quarry, a pond behind an embankment).

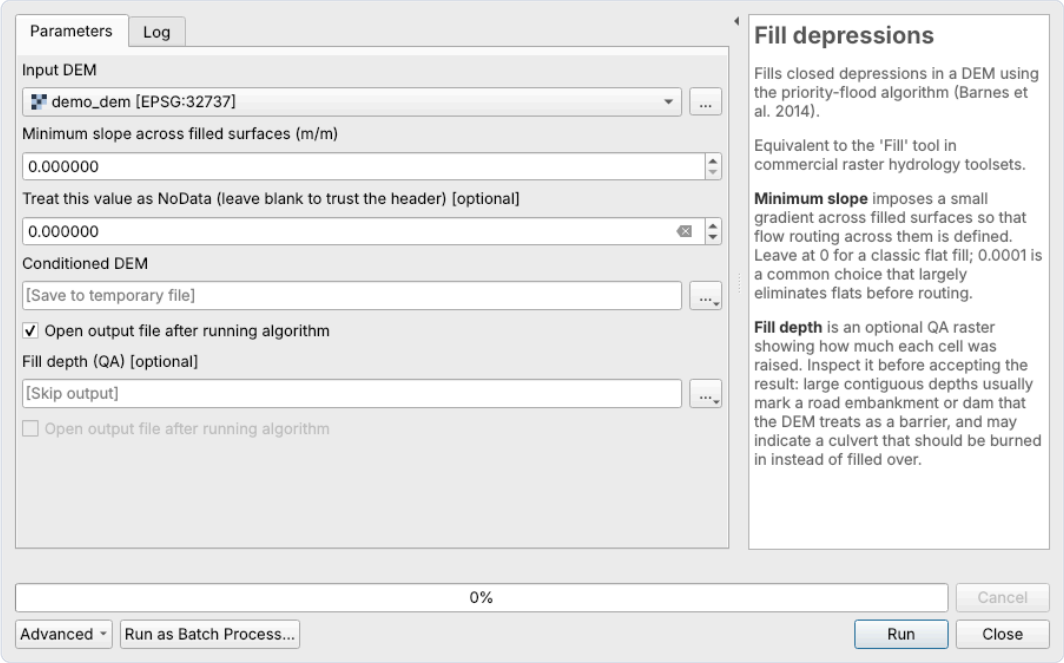


Figure 8. Fill depressions, with the NoData override set for the demonstration DEM.

- **Minimum slope across filled surfaces** (0): leave at 0 for a classic fill. 0.0001 adds a small gradient across filled flats so that routing has a direction, which largely removes the flat-method question.
- **Fill depth (QA)**: an optional raster of how much each cell was raised. A long, deep fill along a road is an embankment the DEM treats as a dam: burn the crossings (section 15) rather than accept the pond.

D8 flow direction

Each cell drains to the neighbour of steepest descent, drop divided by distance (the diagonal distance for diagonal neighbours). Ties are broken by a fixed order (S, W, N, E, then the diagonals), so the result is deterministic. Directions are written in the standard D8 codes (E = 1, SE = 2, S = 4, SW = 8, W = 16, NW = 32, N = 64, NE = 128; 0 = sink or edge outlet), so a grid from another GIS can be used in QEHT's accumulation and catchment tools, and the reverse.

Flats: the one choice that changes catchments

After filling, flat areas have no downhill neighbour, and the flat method decides where they drain:

- **Barnes 2014 convergent** (default since 0.13): flow drains away from higher ground and towards the flat's outlet, so it converges as on real terrain. QEHT reproduces Barnes' own RichDEM implementation cell for cell.
- **Toward lower terrain**: each flat cell takes the shortest path to the outlet. Simpler, and it produces parallel flow lines across large flats.

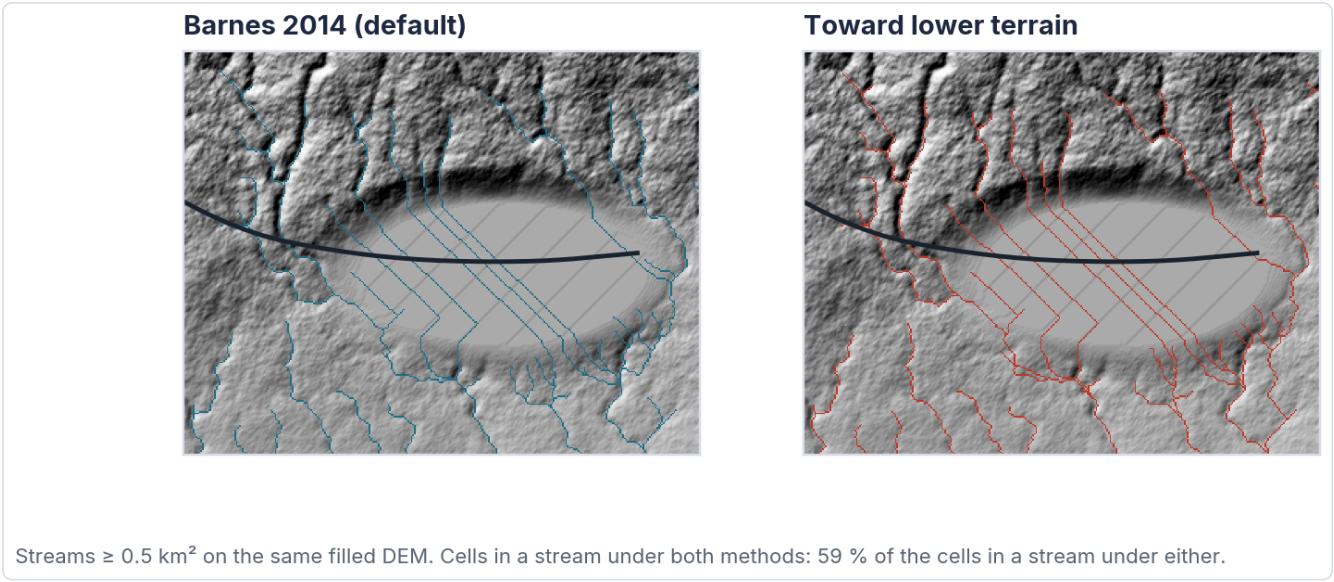


Figure 9. Streams of at least 0.5 km² on the demonstration lake bed under both methods, on the same filled DEM. Off the lake bed they agree; on it they share only 59 % of their stream cells. On a whole-metre DEM (ALOS) a flat plain is mostly flat cells and the method decides the network; on a floating-point DEM (FABDEM) flats are about 1 % of cells and it matters little.

On 80 random 15 km areas across Kenya, each on ALOS AW3D30 and FABDEM, Barnes agreed best with the independent TauDEM (stream F1 0.969 against 0.948 for toward-lower; 0.870 against 0.733 on whole-metre flats), which is why it is the default. `BENCHMARK.md` in the repository has the design and the tables.

> **CHECK FLAT-SENSITIVE CROSSINGS**
> Catchment and flow path characteristics, the package and the pipeline route every crossing under both methods and flag `flat_sensitive = 1` where the area differs by more than 10 %. A flagged catchment is decided by flats or ties, not terrain: check it against a map, imagery or the site before using it.

12 Accumulation and streams

Flow accumulation counts the cells upstream of each cell (a ridge cell reads 0, the standard D8 convention), or their area, or a weight raster. It is a single topological pass with no recursion, so it runs on any grid that fits in memory. Cells in a cycle, which a valid drainage tree never has, are reported rather than dropped.

Stream network and Strahler order thresholds accumulation by cell count or contributing area and orders the network by Strahler's rules on the stream cells only. **Stream network to polylines** traces the raster network into reaches between sources, junctions and outlets, each with `order`, `length`, `drop`, `slope` and `head_type`.

> **THRESHOLD**
> Give the threshold as an area in engineering work: it means the same thing on a 30 m and a 12.5 m DEM. 0.5–1 km² suits road drainage at 30 m; much smaller areas are below what a 30 m DEM resolves.

13 Catchments and flow paths

Delineate catchments from pour points, **Longest flow path** and **Catchment and flow path characteristics** take a pour-point layer. Each pour point is first snapped to the **nearest** stream cell within the *Snap radius* (5 cells). The stream for snapping is the *Snap-to-stream threshold* (200 cells by default): set it low enough to include the tributary a crossing sits on. Snapping to the cell of largest accumulation instead pulls road crossings onto the trunk stream next to them: on the validation site that inflated one catchment forty times.

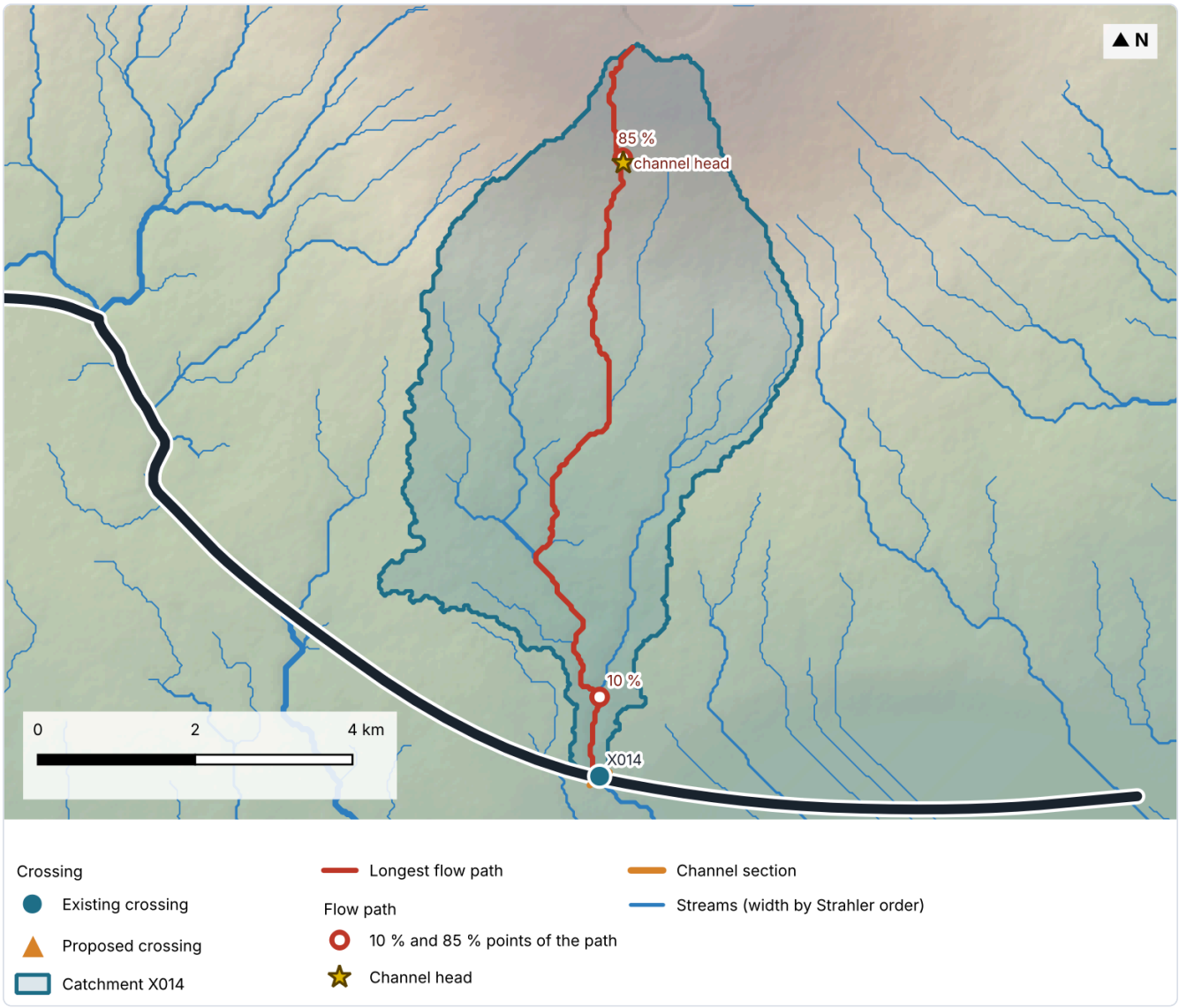


Figure 10. Crossing X014 (27 km²): its catchment, the longest flow path from the divide on the volcano to the road, the 10 % and 85 % points of the path measured from the outlet, the channel head where the path reaches the stream threshold, and the channel section 30 m downstream.

outlet_uid

Every catchment and flow path carries `outlet_uid`, assigned once per run: from a pour-point attribute you choose (existing culvert numbers, for example), or sequential with your prefix (X001, X002 …) along the chainage or downstream first. The three layers of a run share it, so they join on it. `outlet_id`, the feature id of the pour point, changes when a layer is edited or saved again: never join on it.

Four slopes, never merged

Field	Definition	Use
`catch_slope_horn`	Mean terrain gradient over the catchment, Horn 3 × 3	Runoff coefficient and curve-number tables; SCS lag
`catch_relief_ratio`	Relief ÷ longest flow-path length	A basin-steepness index; not a Tc input
`lfp_slope`	Drop ÷ length along the longest flow path	Tc methods that take the whole-path slope
`lfp_slope_1085`	$(z_{85} - z_{10}) \div (L_{85} - L_{10})$, with L_{10} and L_{85} at 10 % and 85 % of the length from the outlet	Kirpich, Bransby-Williams, TRRL and other UK methods

Both catchment slopes are commonly called "catchment slope" and they are not interchangeable: on the validation catchments they differ by a factor of 1.2 to 4.5. The 10–85 slope leaves out the flat bottom 10 % and the steep top 15 % of

the path; `lfp_L10_m`, `lfp_L85_m`, `lfp_z10_m` and `lfp_z85_m` let you check it by hand. **State in the design report which slope you used.**

Flow-path segments and shape

The longest flow path is split at the channel head, the first cell that reaches the stream threshold: `lfp_overland_m` and `lfp_channel_m` with their slopes (they add up to `lfp_length_m`), and the overland part as sheet flow up to a cap (100 m) plus shallow concentrated flow. These are the inputs of the Kerby and TR-55 segment methods. Shape and network indices (perimeter, form factor, elongation and circularity ratios, drainage density, stream frequency, highest Strahler order) are given for information.

> **RAW DEM FOR ELEVATIONS**
> Routing uses the conditioned DEM; reported elevations and slopes come from the raw one, or heights inside filled depressions read as the fill surface. Give the *Raw DEM* to Catchment characteristics; the package and the pipeline always use it.

PART D

Road drainage

From a road centreline to a reviewed set of crossings: where streams cross the road, where water ponds against it, what the channel looks like downstream, whether the DEM agrees with the map, and how far DEM error can move each catchment.

14 Road crossing candidates

Road crossing candidates intersects every D8 flow link of the stream network near the road with the centreline. These are the same segments QEHT's stream polylines are made of, so no crossing is lost to a digitising gap between a stream line and the road.

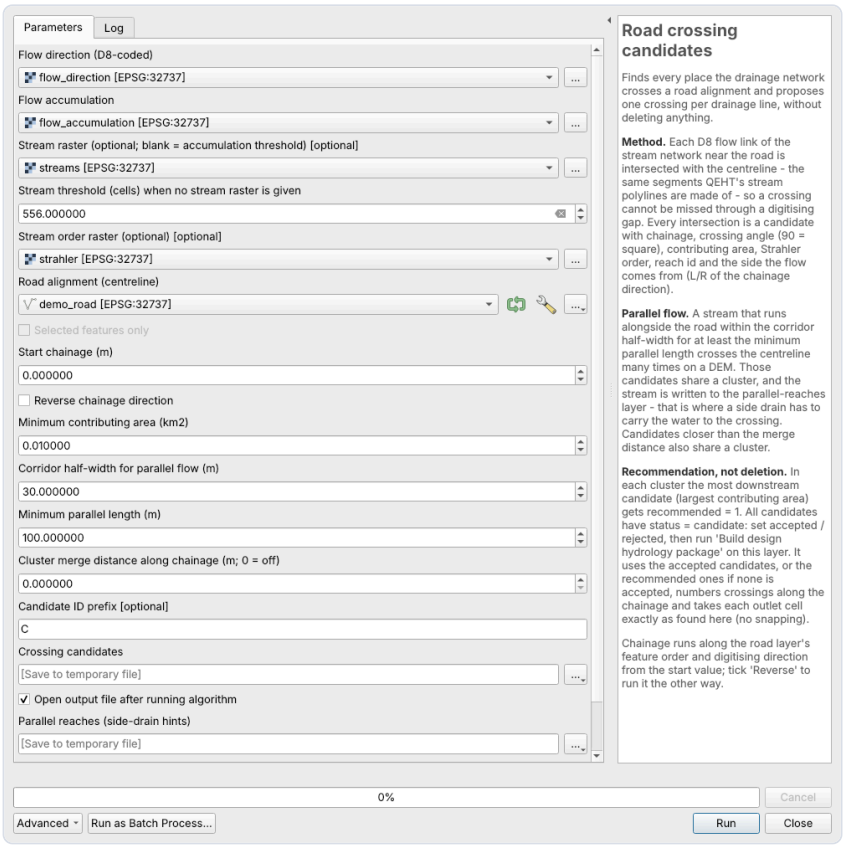


Figure 11. Road crossing candidates, with the rasters of the demonstration run. The stream threshold is 556 cells (0.5 km² on a 30 m grid).

Each candidate carries its chainage, the crossing angle (90 = square), the contributing area, the Strahler order, the reach id, the side the flow comes from (L or R of the chainage direction) and `status = candidate`. Nothing is deleted.

Parallel flow and clusters

A stream that runs beside the road inside the *Corridor half-width* (30 m) for at least the *Minimum parallel length* (100 m) crosses the centreline many times on a DEM. Those candidates share a cluster, the stream is written to the **parallel reaches** layer (where a side drain has to carry the water to the crossing), and the most downstream candidate of the cluster, the one

with the largest area, gets `recommended = 1`. Candidates closer than the *Cluster merge distance* also share a cluster (0 = off).

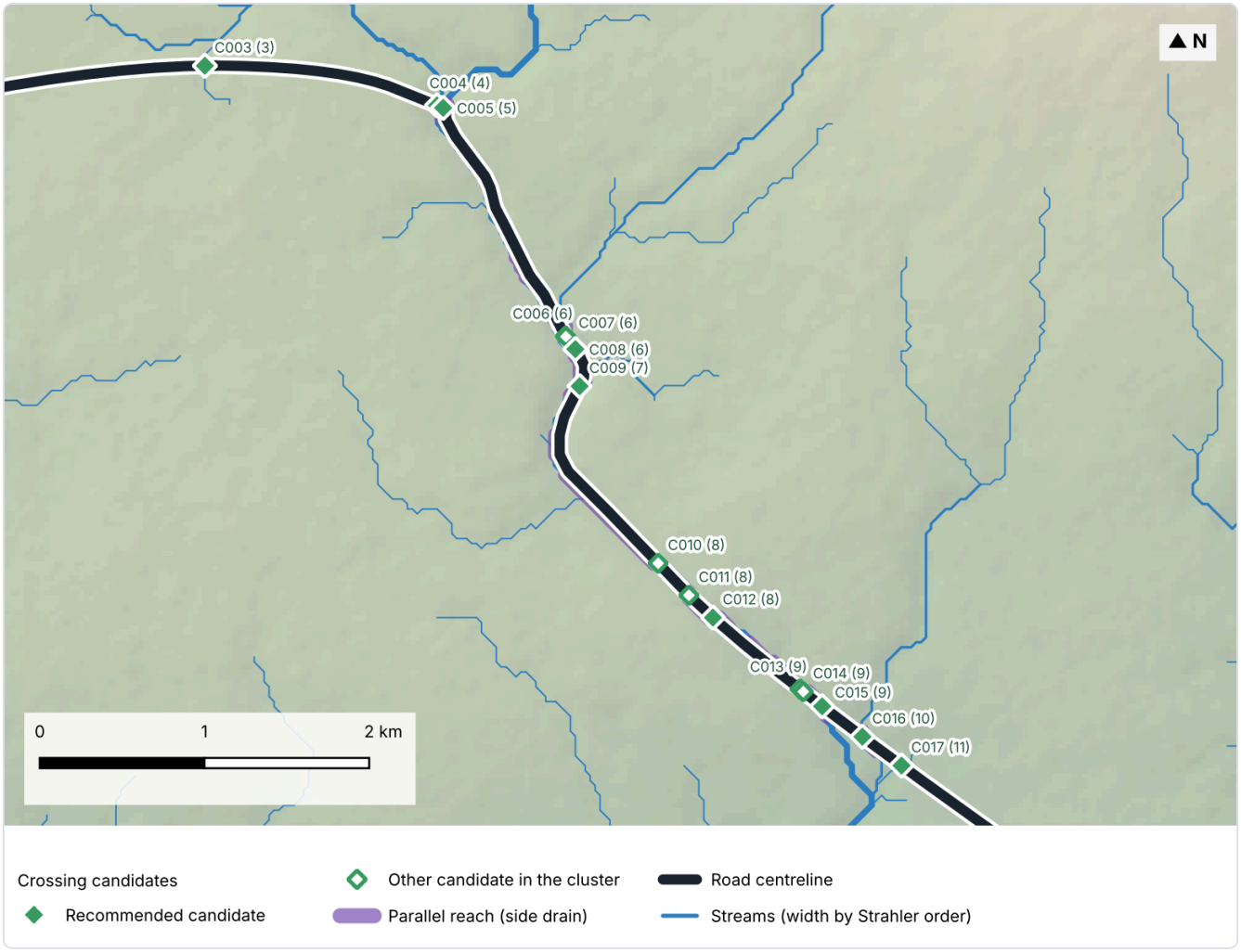


Figure 12. Candidates where the demonstration road runs beside a main valley (km 5.5–11), labelled with their cluster. The stream weaves across the centreline: clusters 6, 8 and 9 each hold several intersections of one stream, and only the most downstream one is recommended. Over the whole road 26 candidates form 20 clusters; 12 candidates lie on parallel reaches.

A crossing used for a cluster covers the cluster's other intersections: the coverage check (section 17) does not propose crossings on them.

15 Burning crossings through embankments

A DSM, a LiDAR or survey DEM, or a fine DEM of a built road shows the embankment as a dam, and filling turns the valley behind it into a lake. **Burn crossings through embankments** cuts a straight breach across the road at each crossing:

- The breach runs perpendicular to the road between two points a *half-length* (30 m) either side of the centreline. Each end moves to the lowest cell within the *search radius* (2 cells) on its own side.
- Cells on the line are lowered to a straight grade from the upstream to the downstream end (at least 0.0001 so it drains). Nothing is raised and nothing outside the breach changes.
- The breach log lists every crossing with cells cut, maximum cut and volume; a crossing with no ground above the grade (no embankment in the DEM) is logged with 0 cells. Give the log to Build design hydrology package and it is stored in the package.

Then fill, route and accumulate the burned DEM. The pipeline does this when *Burn crossings through embankments* is ticked. On a 30 m global DEM the road is usually not in the terrain, and burning changes nothing; check the fill depth along the road first.

16 Alignment ground profile

Alignment ground profile stations the centreline every 10 m: the DEM ground (bilinear), the filled-DEM level and the ponding depth the fill removed, the contributing area (the largest within one cell), the stream crossings with their Strahler order and the longitudinal ground slope. The levels are provisional before the geometric design exists, and a check on DEM height bias afterwards. The package writes the same profile as `alignment_profile`.

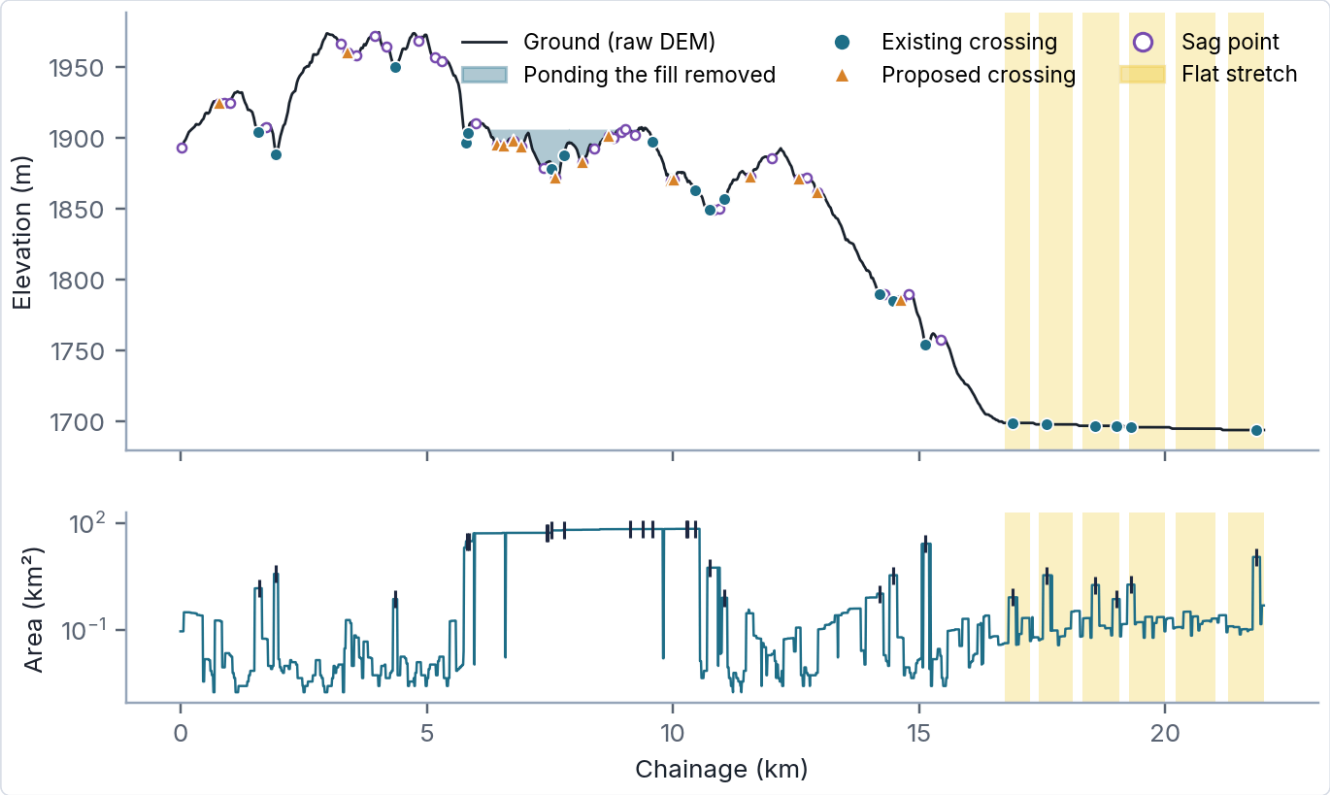


Figure 13. The demonstration road's ground profile with its crossings, sags and flat stretches. Below: the area draining to each station (log scale), with ticks at stream crossings. Between km 6 and 10 the road runs beside a 70 km² stream; from km 16.7 it crosses the flat lake bed. The tool can also write this as a PNG chart.

17 Drainage coverage check

With a road, Build design hydrology package and the pipeline check the road for drainage with no crossing, from the DEM alone; **Drainage coverage check along a road** gives the same findings without delineating anything.

Finding	Rule	What QEHT does
Missing crossing	A stream of at least the stream-threshold area crosses the centreline with no crossing (or crossing of the same cluster) within 50 m of chainage.	Proposes a crossing on the stream (P001 …), delineated and characterised like the existing ones.
Sag point	A low point of the profile (smoothed over 30 m) at least 0.3 m deep, with no stream or crossing within 50 m. The water ponding against the embankment is measured by routing with the centreline as a wall.	Proposes a crossing (equaliser or relief culvert) where the ponding area is at least 0.05 km².
Flat stretch	At least 300 m flatter than 0.5 % along the road and across it over 100 m.	Lists it. Relief culverts for sheet flow may be needed; QEHT does not place or size them.
Mapped river, no crossing	With mapped waterways (section 19): a mapped river crosses the road with no crossing within 50 m.	Lists it; the designer decides.
Small area	A crossing drains less than 0.01 km².	Lists it: redundant or misplaced?

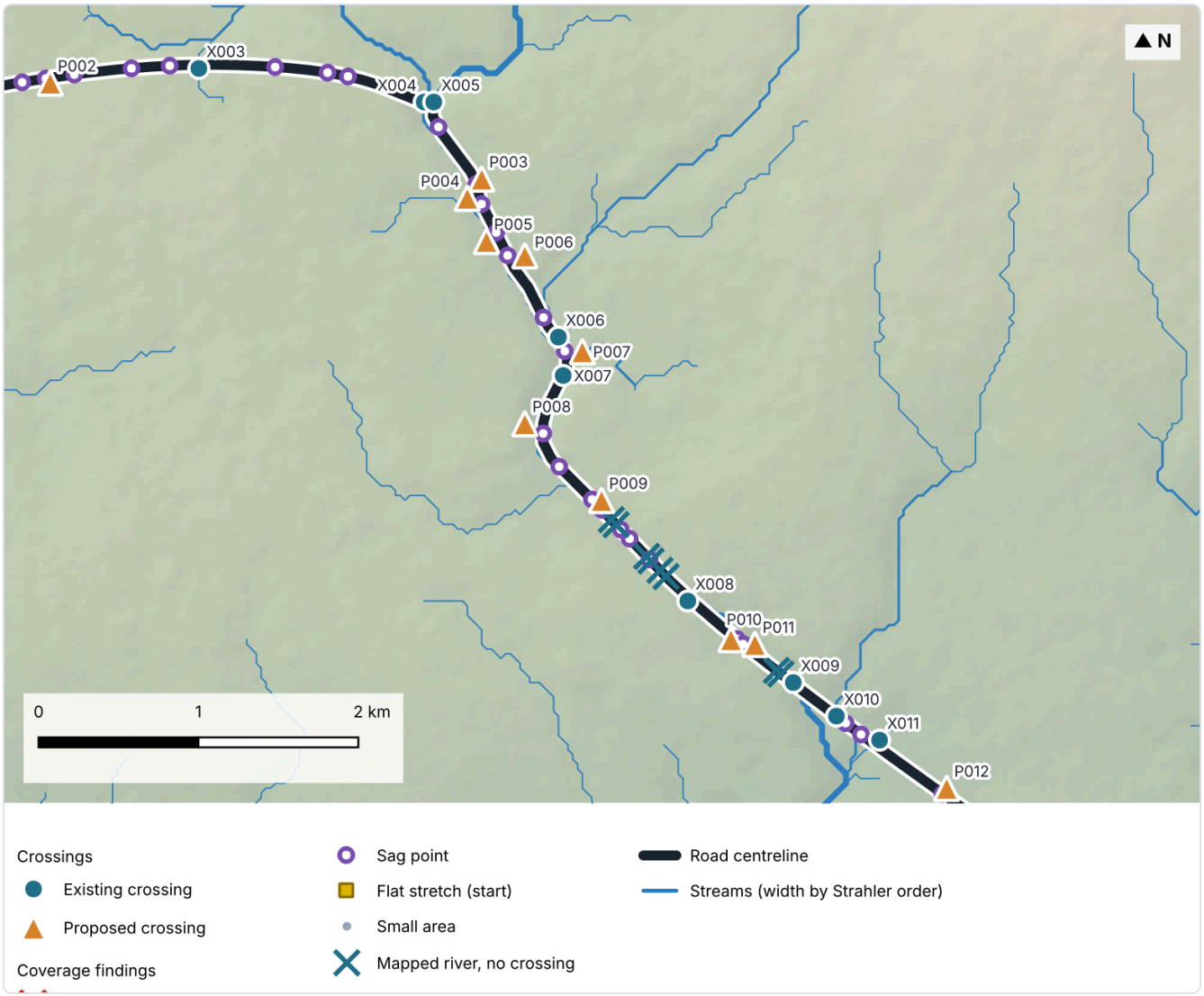


Figure 14. Coverage findings beside the main valley. Every stream crossing is covered (the clusters of Figure 12 cover their stream); the proposed crossings (orange) are sags where water ponds against the road. Where the mapped river crosses the road but the DEM stream runs beside it, the finding is listed for the designer.

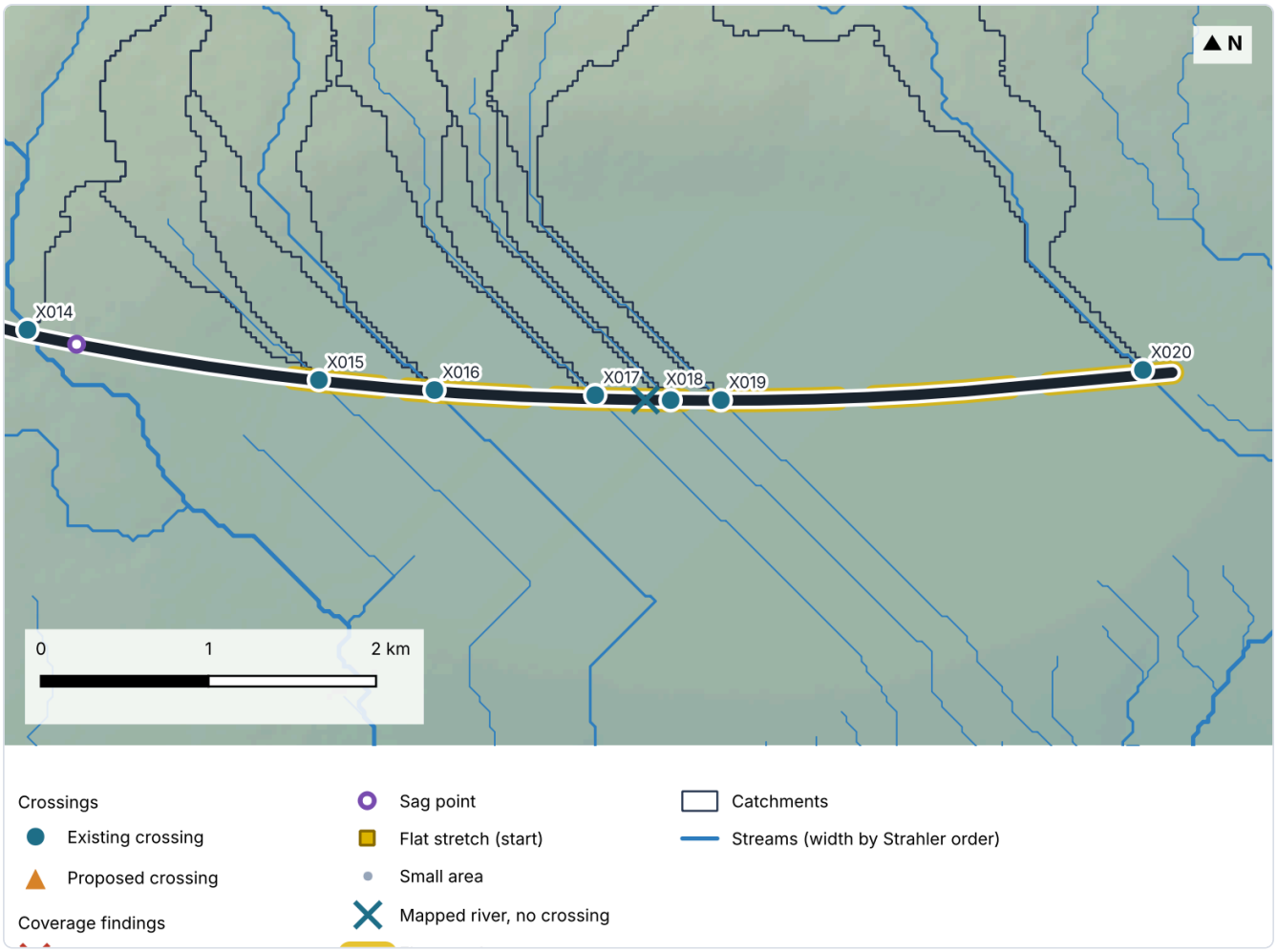


Figure 15. On the lake bed the check finds six flat stretches totalling 4.2 km, where sheet flow needs relief culverts. The catchments here are long strips decided by the flat method, and the parallel streams are typical of flats in a whole-metre DEM.

Adopting proposed crossings. Set a proposed crossing's `status` to `existing` to adopt it, delete those you do not want, then run **Renumber and relink** (section 29): adopted crossings join the X sequence, those still proposed keep the P prefix.

18 Channel sections and floodplain width

Channel section downstream of each crossing

The package and the pipeline cut a cross-section of the natural channel 30 m downstream of every crossing, for the tailwater rating, the waterway and the sediment-continuity check. It lies on the D8 path, perpendicular to the flow (taken over two cells up and down, which smooths D8's 45° steps), sampled bilinearly every half cell over ±150 m, positive on the right looking downstream.

- `xs_bed_m`: the lowest ground within one cell of the centre.
- `xs_bankfull_w_m`, `xs_bankfull_d_m`: bank-full width and depth by break of slope. A bank top is where the side slope falls below 1:20, or to less than half the slope behind; the lower bank governs.
- `xs_w_0p5_m`, `xs_w_1p0_m`, `xs_w_2p0_m`: widths below bed + 0.5, 1 and 2 m, which hold when there is no clear bank; side slopes as H:V; the transect itself as `xs_station_elev_json` and the layer `xs_transects`.
- `xs_quality`: *low* for a bank-full width under 3 cells or no bank on either side, *medium* for one bank or a transect cut by NoData, *high* otherwise; `xs_note` says why.

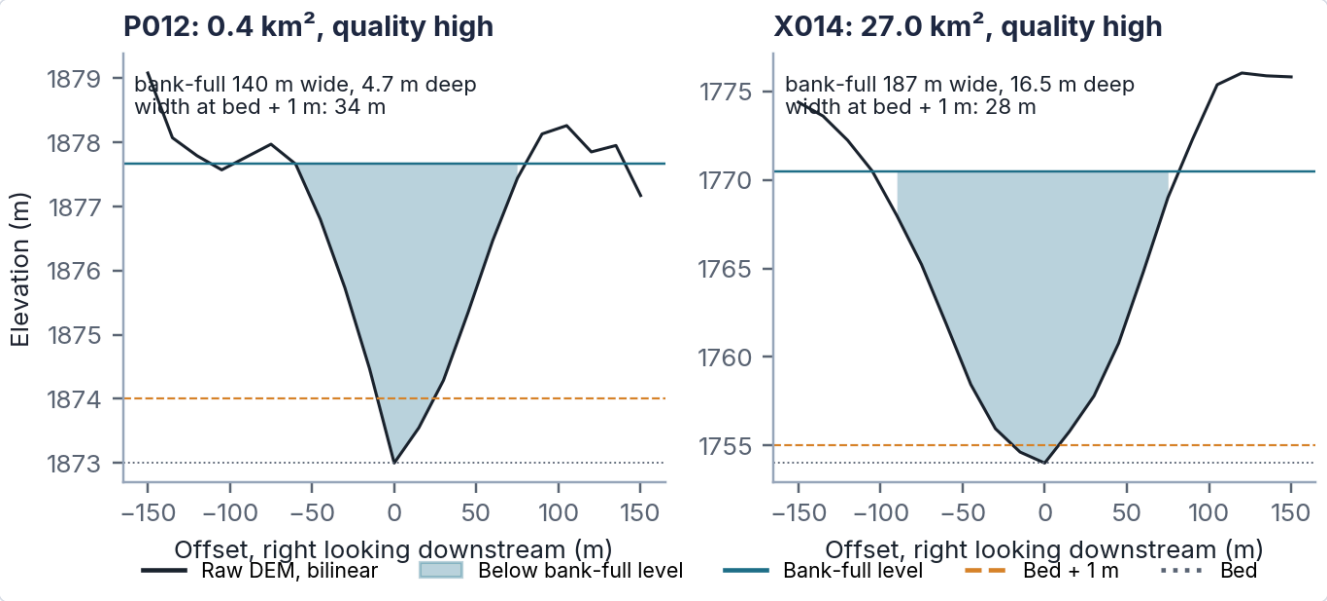


Figure 16. Two sections from the demonstration run. P012 finds a small valley. X014 finds a bank-full depth of 16 m: on a 30 m DEM the break of slope is the valley shoulder, not the channel bank. A bank-full depth of several metres says so. Use survey where it exists; the width at bed + 1 m is the more robust figure.

Floodplain width at large crossings

At every crossing of at least 10 km², QEHT measures how wide the valley floor is along the road: the run of the ground profile around the lowest ground within 50 m of the crossing where the ground is below bed + 0.5, 1 and 2 m (`fp_w_0p5_m` ...), and the same by height above nearest drainage (`fp_hand_w_*`). A width that runs off the end of the profile is a lower bound and `fp_note` says so. The indicator informs the split of the check flood between the main structure and relief culverts; it is not a flood level.

19 Check against mapped drainage

Did the DEM put the river where the map does? Give *Mapped waterways* (OpenStreetMap waterways, HydroRIVERS or a national layer, with an optional name field). QEHT compares them with the DEM streams at a *comparison threshold* matched to the map's scale (1 km² by default; about 10 km² for HydroRIVERS) within a tolerance of 60 m.

- **Network agreement**: precision (share of DEM stream length on the map), recall (share of mapped length found by the DEM) and F1, overall and per catchment.
- **Per crossing**: distance to the nearest mapped waterway, its name and `map_agrees` (1 within the tolerance, 0 up to 1 km, empty beyond or for crossings below the comparison threshold).
- **Divergence reaches**: DEM stream runs of at least 10 cells that leave the mapped course, with their length and the area flowing down them. A large area here is how a catchment that switches between crossings shows up.

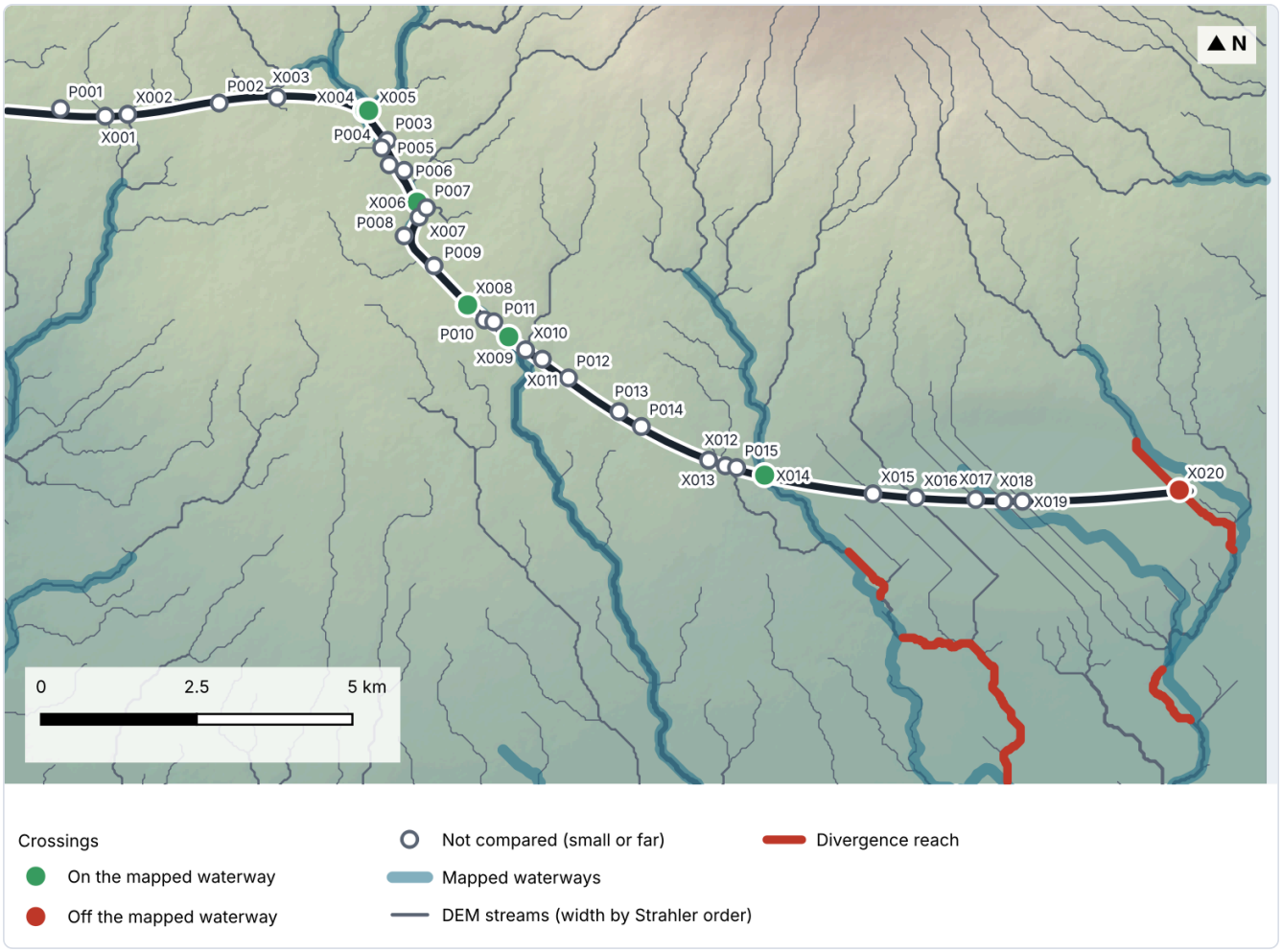


Figure 17. The demonstration's mapped rivers (blue, at least 8 km²) against the DEM streams. Overall precision 0.79, recall 0.73, F1 0.76. Six large crossings sit on a mapped waterway; X020, at the end of the lake bed, is 322 m from it. The five divergence reaches (red, 9.9 km) are where the DEM sends the river another way across the flats.

20 Large DEMs: clip to the contributing area

A national mosaic will not fit in a laptop's memory, and clipping by hand can cut catchments. **Clip DEM to the road's contributing area** (or the pipeline option) finds the area that drains to the road on a coarse copy and reads only that window at full resolution:

1. The coarse copy takes the **minimum** of each block, so valleys are kept; the block size keeps the copy under 4 million cells.
2. The copy is filled and routed; every coarse cell that drains to the road buffer (200 m) is kept.
3. That area is widened by the margin (1,000 m plus two coarse cells), and its bounding box (or the mask) is read from the full DEM.

The tool reports the memory estimate before and after. The clipped DEM is tagged, and the package then gives every catchment `clip_edge` : 1 when it touches the edge of the clip, with a warning to enlarge the margin.

21 DEM uncertainty at crossings

One catchment area from one DEM hides how far the DEM's error can move it. **DEM uncertainty at crossings** (or the package and pipeline option) runs a Monte Carlo:

1. A spatially correlated Gaussian error is added to the raw DEM: standard deviation σ, correlation $\exp(-r^2/L^2)$ with L = 90 m by default.

2. The DEM is filled, routed and accumulated again, and each crossing's pour point is snapped to the stream.
3. Area, longest flow path, 10–85 slope and Kirpich Tc are measured. This repeats N = 50 times (seed 2026, recorded, so the run repeats exactly).

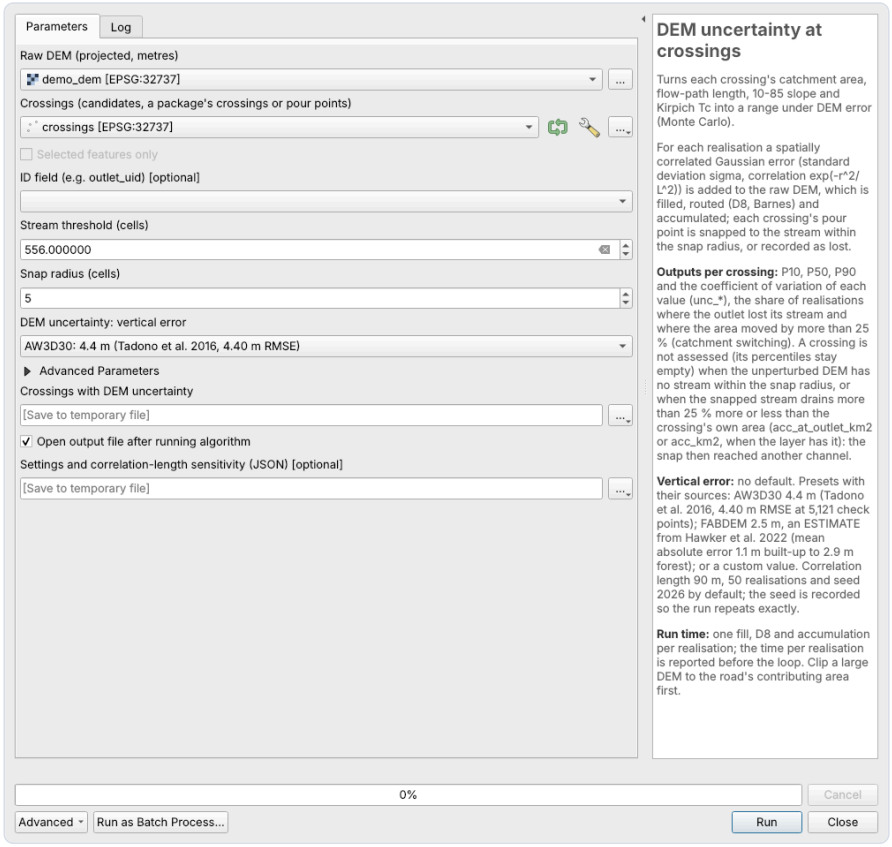


Figure 18. DEM uncertainty at crossings, on the crossings of the demonstration run with the AW3D30 preset.

Vertical error, with no default. AW3D30: 4.4 m, the RMSE of the JAXA validation (Tadono et al. 2016). FABDEM: 2.5 m, an ESTIMATE from the mean absolute errors of Hawker et al. (2022), 1.1 m built-up to 2.9 m forest. Or your own value.

Per crossing the run gives P10, P50, P90 and the coefficient of variation of each value (`unc_*`), the share of runs where the outlet lost its stream (`unc_lost_pct`) and the share where the area moved by more than 25 % (`unc_switch_pct`), the sign of a catchment that switches between crossings. For the three largest crossings the run is repeated at 0.5 × and 2 × the correlation length. A crossing is not assessed when the unperturbed DEM has no stream within the snap radius, or when the snap reaches another channel (the snapped area differs by more than 25 % from the crossing's own): proposed sag crossings below the stream threshold are the usual case.

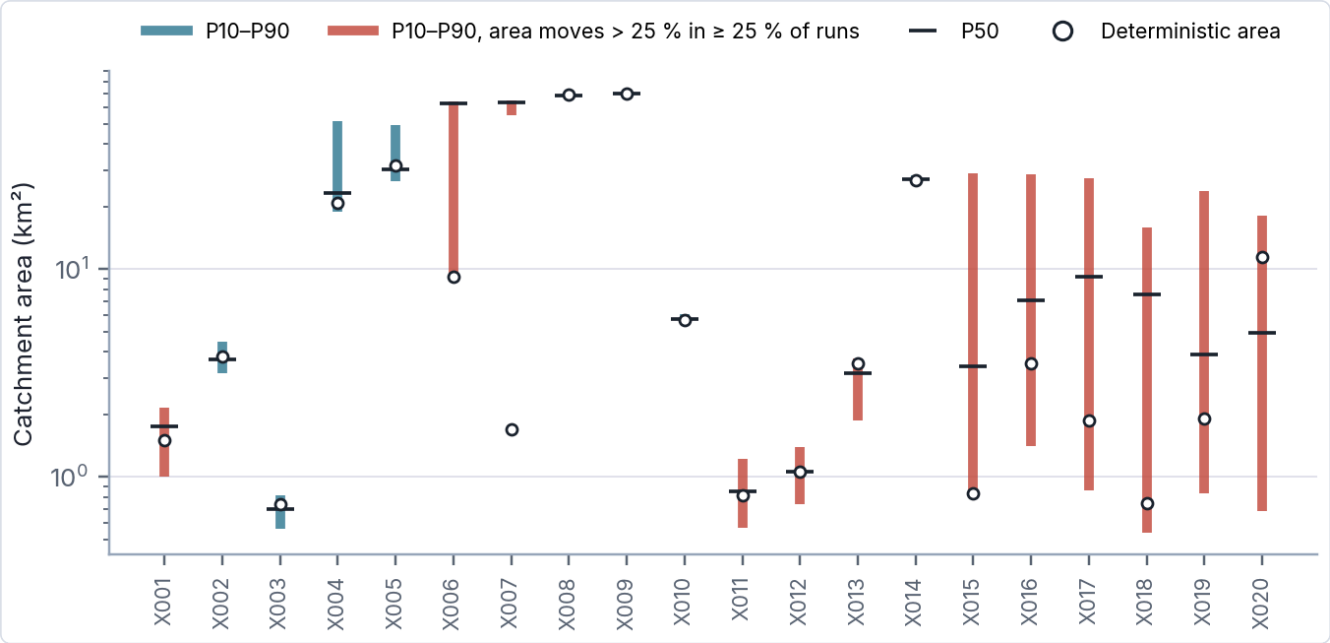


Figure 19. Area ranges at the existing crossings, 30 realisations with σ = 4.4 m. Crossings on steep, well-defined valleys (X008, X009, X014) barely move. In the valley beside the road X006 and X007 can capture the main stream, and on the lake bed every catchment switches in about half the runs: their design areas are a choice the DEM cannot make.

The time per realisation is reported before the loop (under 1 s for the 0.55-million-cell demonstration DEM). Clip a large DEM first.

PART E

Soils, runoff and erosion

The catchment inputs of a design-flood and sediment assessment. Each block is computed when its input is given, and every value says where it came from.

22 Soils and USLE K

Soil, rainfall and runoff parameters for catchments (and the soil inputs of the package and the pipeline) adds a soil block to every catchment: area-weighted topsoil sand, silt, clay, organic carbon, coarse fragments and bulk density; USDA texture; dominant FAO drainage class; USLE K; a hydrologic soil group; and the share of the catchment covered by soil data.

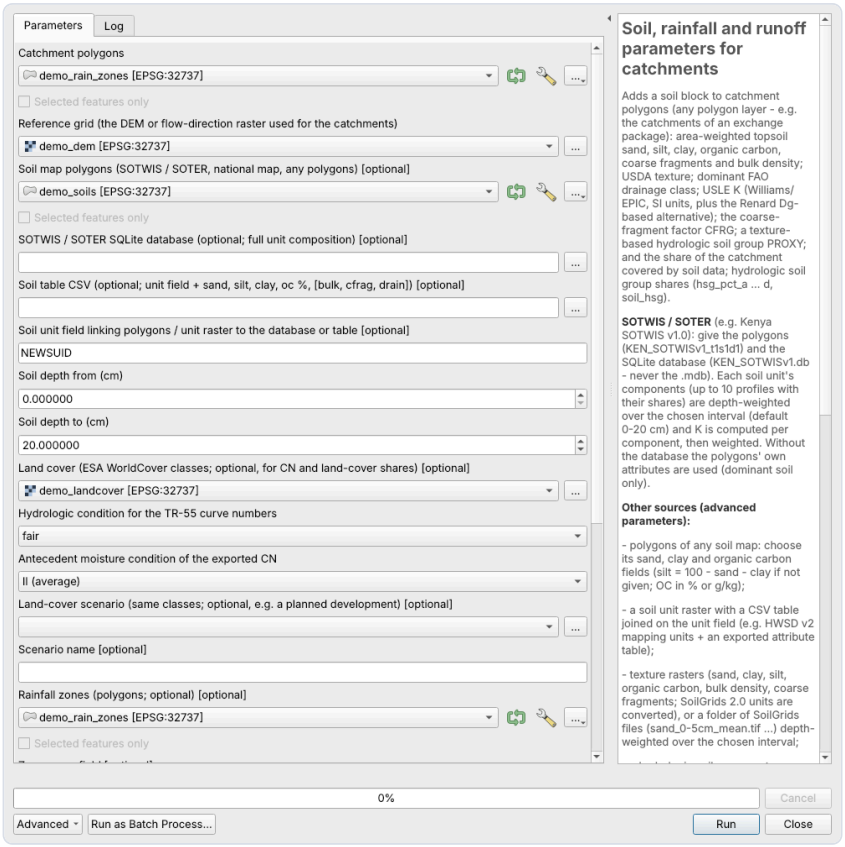


Figure 20. Soil, rainfall and runoff parameters for catchments, with the demonstration soils, land cover and rainfall.

Soil sources

Every source becomes the same per-cell description on the DEM grid, so the catchment fields mean the same thing whatever the source:

Source	Give
SOTWIS / SOTER database	The soil polygons and the SQLite database. Each unit's components (up to ten profiles with their shares) are depth-weighted over the chosen interval (0–20 cm by default); K is computed per component, then weighted.
Any soil polygon map	Choose its sand, clay and organic-carbon fields (OC in % or g/kg; silt = 100 − sand − clay when not given), plus optional bulk density, coarse fragments, drainage, HSG and K fields.
Polygons + soil table	Polygons with a unit code and a CSV with the unit code and `sand`, `silt`, `clay`, `oc` [+ `bulk`, `cfrag`, `drain`].
Soil unit raster + table	For example HWSD v2 mapping units with an exported attribute table.
Texture rasters or SoilGrids	Sand, clay (silt) and OC rasters, or a folder of SoilGrids 2.0 files (`sand_0-5cm_mean.tif` ...), converted from SoilGrids units and depth-weighted.
HSG and K overrides	A hydrologic soil group raster (HYSOGs250m codes; dual groups A/D, B/D, C/D count as D) or field, and a K raster or field, which replace the texture-based values where they have data.

K and the hydrologic soil group

K is the Williams (EPIC) relation in SI units (t·ha·h/(ha·MJ·mm)), with the Renard Dg-based value as an alternative and the coarse-fragment factor reported separately. Because the relation is non-linear, K is computed per component or per cell and then averaged, never from averaged texture. The hydrologic soil group, when no HSG data is given, is a PROXY from texture and drainage (A: sand, loamy sand; B: sandy loam, loam, silt loam; C: sandy clay loam; D: clay loams and clays; poorly drained soils move to D). `soil_hsg_source` and `usle_k_source` say which was used.

> **TREAT K AS AN ESTIMATE**
> Across Kenya, SOTWIS-derived K has a median of 0.029 against 0.022–0.024 for the ESDAC global K rasters over the same cells, with a correlation of only 0.24–0.40. The two are independent estimates at very different resolutions. Report which one you used.

23 Curve number, Rational C and scenarios

With a land-cover raster in ESA WorldCover classes, each cell's land cover and hydrologic soil group give a curve number from USDA TR-55 Table 2-2, for the chosen condition (fair, good or poor). The catchment value `cn_ii` is the cell mean, exported at the antecedent moisture condition you choose (`cn_export`, AMC I, II or III).

- The match from WorldCover classes to TR-55 covers (tree → woods, shrub → brush, grass → pasture, crop → row crops, built-up → commercial 85 % impervious, bare → fallow) is a PROXY, flagged in `runoff_json`. Replace it with your own *Curve number lookup CSV* (`class, A, B, C, D`).
- Rational C comes only from a lookup you supply (*Rational C lookup CSV*); none ships.
- Land-cover shares `lc_pct_*` are always given with a land-cover raster.

Land-cover scenarios

Give a second raster with the same classes (a planned development, a future land-use map) as *Land-cover scenario*, with a name. It runs through the same lookups and the same soil groups as the baseline, so the change comes from land cover only: `cn_ii_scn`, `d_cn`, `rational_c_scn`, `d_rational_c`, scenario land-cover shares, and with an erosion folder the scenario C factor and soil loss (A scaled per cell by C_{scn}/C, which holds because A = R K LS C P).

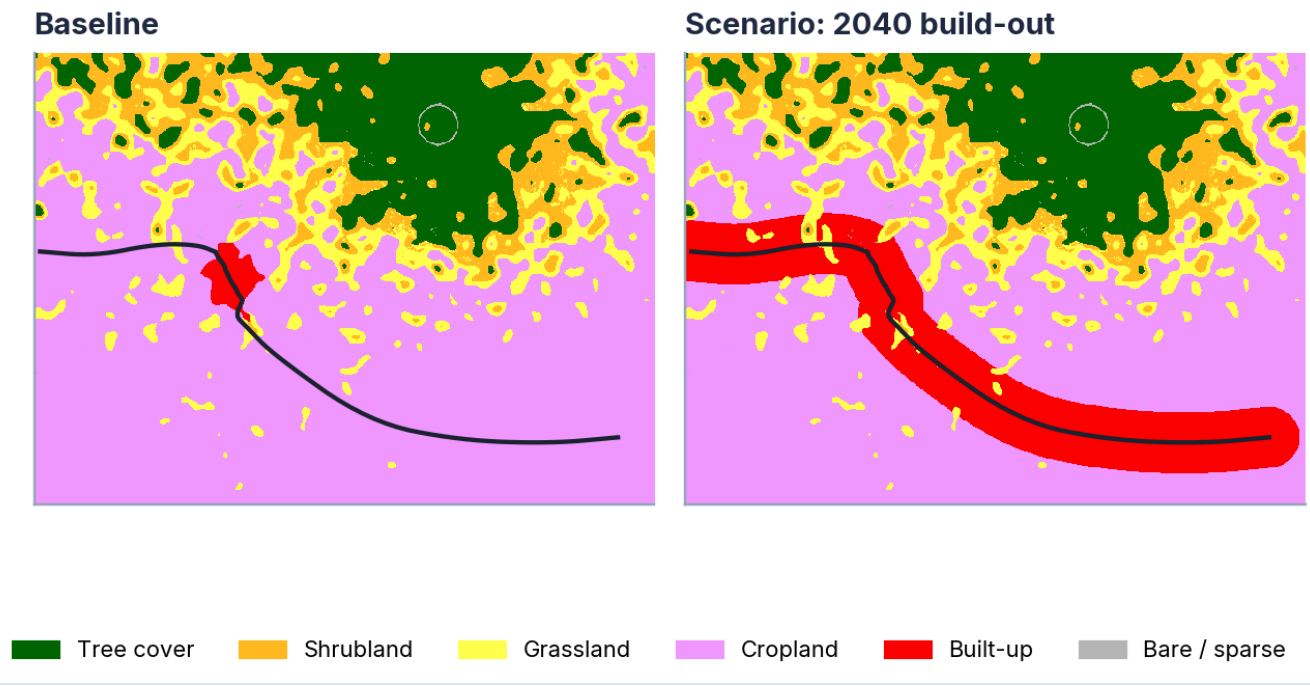


Figure 21. The demonstration's baseline land cover and its 2040 scenario, in which cropland within 1 km of the road becomes built-up. Small catchments beside the road gain up to 6 curve-number points; the large ones, mostly forest and shrub on the volcano, barely change.

Land-cover scenario: 2040 build-out

demo_landcover_2040.tif (resampled near to the DEM grid). Same lookups and soil groups as the baseline, so the change comes from land cover only. A_scn = A x C_scn / C per cell; C_scn from the WorldCover → C lookup; baseline C: ESA WorldCover 2021 class lookup (demo_landcover.tif (resampled near to the DEM grid))

outlet_uid	area_km2	cn_ii	cn_ii_scn	d_cn	ero_a_mean_tha	ero_a_mean_tha_scn	d_ero_a_mean_tha
X001	1.505	87.92	90.583	2.663	61.34	44.604	-16.736
X002	3.785	84.769	86.573	1.803	69.126	54.092	-15.034
X003	0.74	87.838	92.444	4.606	35.278	19.688	-15.59
X004	20.849	73.439	74.365	0.926	56.697	49.317	-7.379
X005	31.708	67.568	67.633	0.0657	46.515	45.891	-0.624
X006	9.219	83.029	83.147	0.118	54.199	53.608	-0.592
X007	1.702	89.047	92.505	3.458	50.662	34.598	-16.064
X008	69.507	73.842	74.448	0.606	54.27	49.537	-4.733
X009	70.448	74.044	74.721	0.676	55.637	49.957	-5.679
X010	5.684	87.617	88.354	0.737	70.23	63.402	-6.828
X011	0.814	89	94.27	5.27	41.097	21.761	-19.336
X012	1.062	88.716	94.375	5.659	47.251	23.865	-23.386
X013	3.502	88.683	93.284	4.601	52.36	30.925	-21.435
X014	26.962	84.499	84.62	0.121	78.186	76.693	-1.493
X015	0.837	89	91.252	2.252	60.589	42.431	-18.157
X016	3.502	88.336	88.609	0.273	88.426	84.065	-4.362
X017	1.859	88.828	89.081	0.253	88.568	87.686	-0.882
X018	0.747	89	89.535	0.535	91.357	91.032	-0.325
X019	1.905	88.5	88.687	0.187	86.487	86.398	-0.0885
X020	11.413	86.069	86.131	0.062	79.364	79.132	-0.232
P001	0.11	89	95	6	30.258	15.129	-15.129
P002	0.156	89	95	6	32.777	16.389	-16.389
P003	0.0576	95	95	0	44.33	44.33	0

Figure 22. The run report's scenario table.

24 Rainfall

- **Rainfall zones**: a polygon layer and its name field. Per catchment the dominant zone (`rain_zone` ; ties go to the name that sorts first, so the result does not depend on layer order), its share and the share of every zone. A warning lists catchments whose dominant zone covers less than 80 %: pick their design zone by hand.
- **Mean annual rainfall**: a raster in mm/yr (a CHIRPS climatology, for example), bilinear to the DEM grid: `map_mm` . Values that cannot be annual millimetres (mm/day, tenths of mm, negative) are refused.
- **R from rainfall** (optional): RUSLE R per cell from the rainfall by Renard and Freimund (1994) or Lo et al. (1985), labelled an ESTIMATE. Neither relation comes from East Africa; an erosivity raster such as GloREDa is better.

25 Time of concentration

The package and the pipeline compute the time of concentration of every crossing by five published methods, side by side. **QEHT does not choose one.**

Field	Method	Inputs
`tc_kirpich_min`	Kirpich (1940): 0.0195 $L^{0.77}$ $S^{-0.385}$	flow-path length, 10–85 slope
`tc_kerby_kirpich_min`	Kerby (1959) overland + Kirpich channel	overland and channel lengths and slopes; retardance N
`tc_scs_lag_min`	SCS / NRCS lag (NEH 630 ch. 15), Tc = lag / 0.6	flow-path length, `cn_ii` , Horn catchment slope
`tc_tr55_min`	TR-55: sheet (≤ 30 m) + shallow concentrated + channel (Manning)	P2, sheet n, overland slope, channel slope, the bank-full section, channel n
`tc_bransby_williams_min`	Bransby-Williams: 14.6 L / ($A^{0.1}$ $S^{0.2}$)	length, area, 10–85 slope

- **Validity flags.** Each method has a `tc_<method>_flag` , *within* or *outside: …*, from its published range: Kirpich 0.004–0.45 km² and 3–10 % slope; SCS lag up to 8 km² and CN 50–95; Kerby overland length up to 365 m; TR-55 sheet flow up to 30 m. Values are reported either way.
- **No invented inputs.** P2 (the 2-year 24-hour rainfall, mm) is your value or raster, with no default; without it TR-55 is empty with the note "give P2". Sheet-flow n and Kerby N are PROXY lookups matched to WorldCover and weighted along the flow path; give your own CSV (`class,value`). Channel n defaults to 0.035 (clean natural channel, Chow 1959).
- `tc_basis_json` holds the inputs each method used, and `tc_params_json` the ranges with their sources, P2 and the lookups.

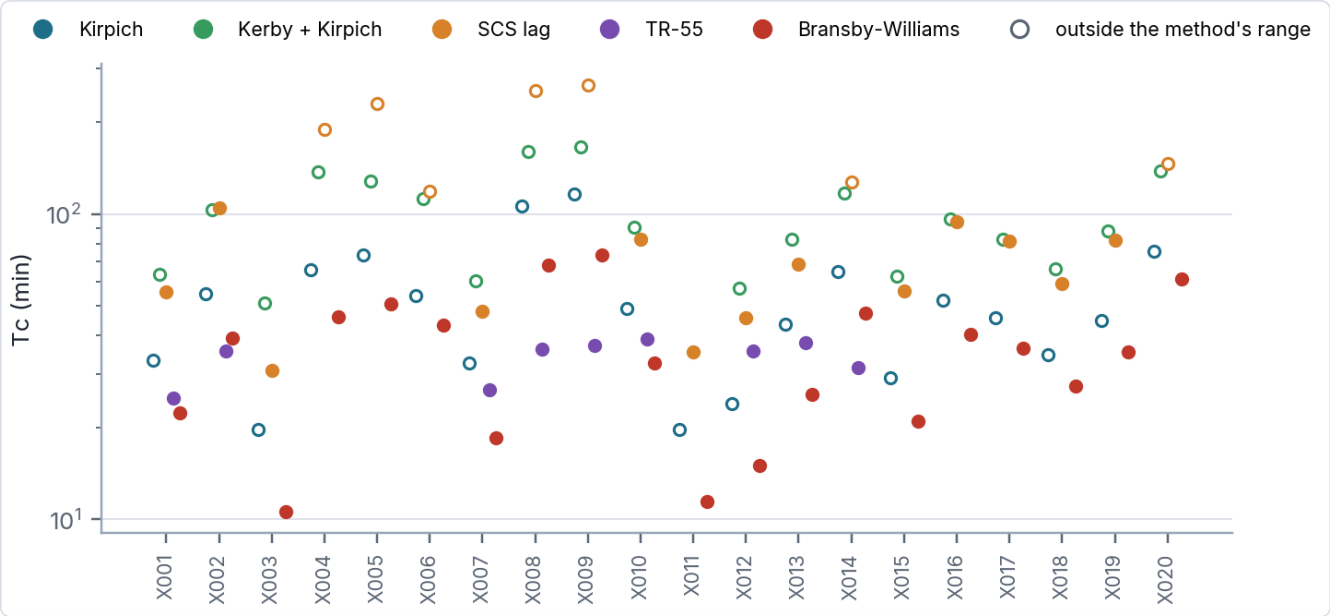


Figure 23. Tc at the demonstration's existing crossings (P2 = 60 mm). Hollow markers are outside the method's published range, which is most of them: Kirpich and SCS lag were derived on small catchments. The spread between methods at one crossing is often a factor of three or more. Choose the method your design standard prescribes and state it.

26 Erosion indices and RUSLE

Erosion indices and RUSLE soil loss writes one folder with the terrain indices, an optional soil-loss raster, severity classes (with colours, class names and a QGIS style) and `erosion_run.json`, which records every factor's source and every setting.

- **Indices** on the raw DEM: Horn slope; specific catchment area A_s = (upslope cells + 1) × cell area / cell size; SPI = A_s tan β; TWI = ln(A_s / tan β); LS by Moore and Burch (m = 0.4, or m by slope class) or Desmet and Govers.
- **RUSLE** A = R K LS C P (t/ha/yr). Each factor is a raster, a single value or a dataset: R from a raster (GloREDa recommended) or the rainfall estimate; K from the soil inputs; C from WorldCover through an editable class lookup, flagged as a PROXY; P = 1 unless given. **If R, K or C is missing, no soil loss is computed**: QEHT never invents a factor and falls back to LS-only classes labelled "terrain potential".
- **Classes.** SPI by percentiles of the extent or fixed ln(SPI) 0 / 5 / 10; RUSLE by 5 / 12 / 25 / 50 or 5 / 10 / 20 / 40 t/ha/yr, or your own breaks; the combined class is the higher of the two unless you give a 5 × 5 matrix. `class_extents.csv` gives the area per class, split into channel and hillslope cells.
- **Sediment transport index**: `sti_overland.tif`, Moore and Wilson (1992) on overland cells (channels NoData), a relative indicator of where slopes deliver sediment. It is not used in soil loss, the classes or the composite.

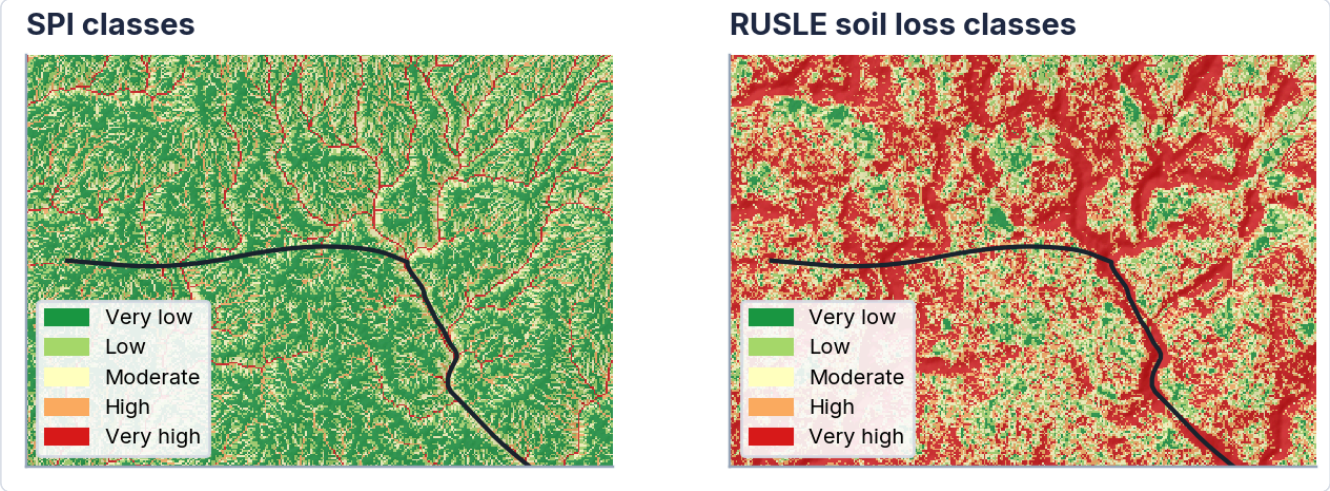


Figure 24. SPI and RUSLE classes around the western half of the demonstration road. SPI picks out the channels; RUSLE is high on the steep cropped slopes. The demonstration factors are invented and the classes illustrate the method only.

With the erosion folder, the package adds an erosion block: per catchment the mean and P90 soil loss, LS, channel ln(SPI), area-weighted K, C and P, class shares, gross soil loss, the sediment delivery ratio (SDR = 0.565 $A^{-0.125}$) and sediment volume; per crossing the ln(SPI) at and above the outlet, local slope, LS and TWI, a deposition indicator on the approach channel, and the hydrodynamic impact score 0.4·SPI + 0.3·RUSLE + 0.3·sediment (Low / Moderate / High / Severe) with an indicative mitigation, following the A14 corridor study (Akello and Omosa 2025).

27 Erosion along the road and side drains

Sample erosion along alignment reads the erosion folder and a road centreline. Stations every 10 m carry the maximum and mean ln(SPI), LS, soil loss and TWI on each side within a 50 m buffer; consecutive stations with the same worst class form reaches whose lengths add up to the road length.

Side-drain siltation. Give the flow direction (and the raw DEM for the flag) and each station also measures the slope that feeds each side drain: the cells sampled 10–50 m out whose D8 path reaches the road strip without entering a channel. `sti_p50_lhs`, `sti_p90_lhs` and the same for the right side come from the overland STI; classes (low, moderate, high, very high) are relative to this corridor's own stations; `siltation_lhs` / `_rhs` = 1 where the class is high or very high and the longitudinal slope is under 1 %. The package writes the reaches as the layer `corridor_sti`.

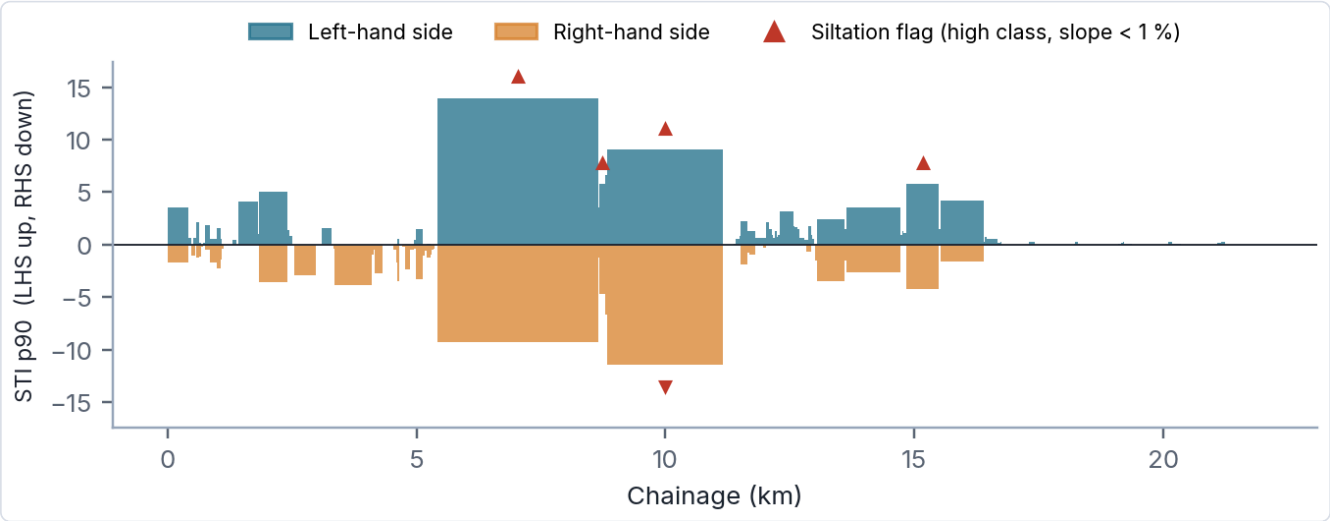


Figure 25. STI P90 of the slopes feeding each side drain along the demonstration road, with the reaches flagged for siltation. The highest values are beside the main valley, where long slopes drain straight onto the road; on the lake bed no slope delivers sediment.

PART F

The design hydrology package

One self-describing GeoPackage per run: crossings, catchments and flow paths linked by a stable key, with the run's provenance and a dictionary of every field inside the file.

28 What the package holds

Build design hydrology package (Interoperability; algorithm id `buildheasexchange`) runs snapping, delineation, the longest flow path and the characteristics for every pour point, adds every optional block, and writes one GeoPackage (schema `qeht-heas-1`). The pipeline uses it internally and keeps it when *Keep the design hydrology package* is ticked.

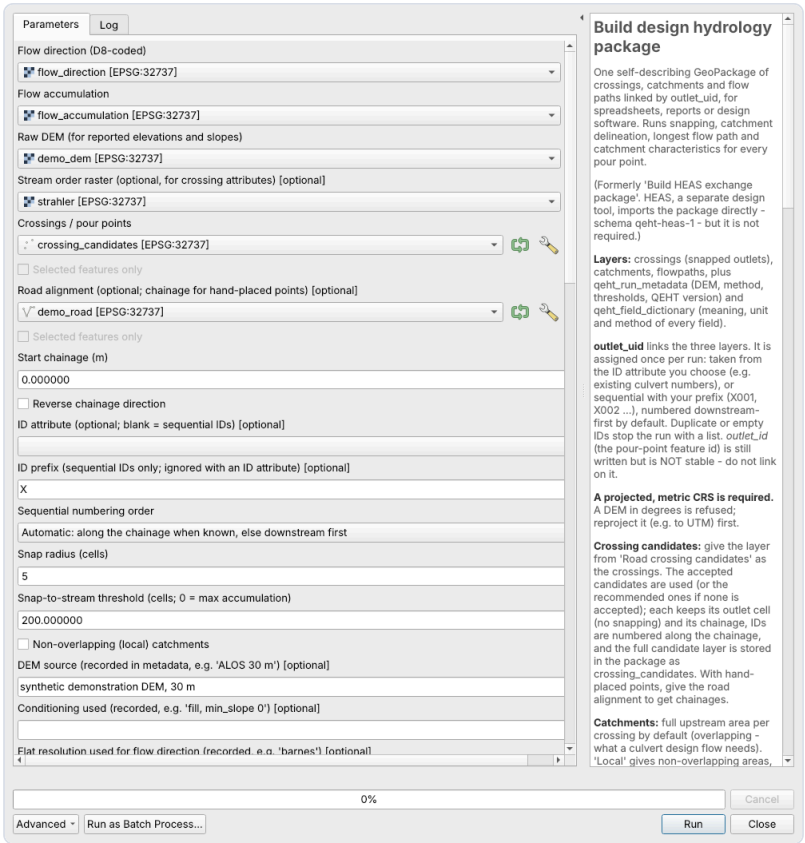


Figure 26. Build design hydrology package, upper part. Further down are the coverage, section, floodplain, soil, runoff, rainfall, erosion, mapped-drainage, uncertainty and Tc parameters, the same as the pipeline's.

Table	Content
`crossings`	Snapped outlets with `outlet_uid`, input and snapped coordinates, snap distance, contributing area, Strahler order, chainage, status, channel slopes, channel section, floodplain, Tc, mapped-drainage, uncertainty and erosion fields
`catchments`	One polygon per crossing with area, elevations, the two catchment slopes, shape, soils, curve number, rainfall, scenario and erosion fields

Table	Content
`flowpaths`	The longest flow path per crossing with length, drop, the two path slopes, the 10–85 points and the segments
`qeht_run_metadata`	QEHT version, run time, DEM path and SHA-256, DEM source and preparation, CRS, cell size, conditioning, flat method, thresholds, snapping, ID scheme, every parameter, and a JSON record per block
`qeht_field_dictionary`	Every field of every layer with its type, unit, meaning, method and downstream use (Appendix B)
Optional layers	`road_alignment`, `alignment_profile`, `crossing_candidates`, `coverage_check`, `sag_points`, `flat_stretches`, `xs_transects`, `corridor_sti`, `mapped_rivers_used`, `drainage_divergence`, `burn_log`, `renumber_log`, `rasters`

Rules

- The three main layers share `outlet_uid`, so any reader links them by key, never by row order or position.
- A projected, metric CRS is required; a DEM in degrees is refused before any computation.
- Empty or duplicate IDs, and two pour points snapping to the same cell, stop the run with a list.
- Missing values are NULL, never 0. `acc_at_outlet_km2`, read from the accumulation raster, equals `area_km2` for full catchments: a built-in check.
- The schema only grows. Fields and layers are added under `qeht-heas-1`; renaming or removing one would be `qeht-heas-2`, and the validator refuses unknown major versions.
- The file is written with the Python standard library, so it is byte-for-byte the same on every platform; the tests check it with GDAL's GeoPackage validator.

IDs. *ID attribute* keeps your own numbering (culvert numbers from an inventory, for example). Without it IDs are sequential with the *ID prefix* (X): along the chainage when the crossings have one, otherwise downstream first (the largest area is 001). Proposed crossings use the *Proposed prefix* (P).

HEAS, a separate hydrology–hydraulics design tool, imports the package directly. It is not required: the package opens in QGIS like any GeoPackage, and the pipeline also writes every layer as CSV.

29 Editing crossings: renumber and relink

After you delete, move or add crossings in a package, or adopt proposed ones, **Renumber and relink exchange package** writes a new package:

1. Unmoved crossings keep their outlet cell; moved and added ones are snapped to the stream.
2. Crossings are ordered by chainage (recomputed from the road for moved and added points) or downstream first.
3. IDs are re-issued gaplessly with the package's prefix; crossings still marked proposed keep the P prefix.
4. Catchments and flow paths are recomputed, and `renumber_log` records every old → new ID with its status (*unchanged*, *renumbered*, *new*, *deleted*) and how far a moved point moved.

IDs change only when you run this tool. Run the report tool on the new package for an updated report.

30 Batch, Modeler and scripts

Batch. Every tool has **Run as Batch Process...**, for example to run the pipeline on several roads.

Graphical Modeler. `examples/qeht_corridor.model3` chains Fill → D8 → accumulation → streams → crossing candidates → Build design hydrology package, with the DEM, road and stream threshold as inputs. Open it with **Model → Open Model...** as a starting point for your own chains.

Python. The tools run from the QGIS Python console or a script:

```
import processing
processing.run("qeht:hydrologypipeline", {
    "DEM": "C:/data/dem_utm37s.tif",
    "ROAD": "C:/data/road.gpkg",
    "STREAM_KM2": 0.5,
    "NODATA_OVERRIDE": 0,
    "OUTPUT": "C:/runs/road_a"})
```

Parameter names are in Appendix A. **Run report from a design hydrology package** and **Export package to KMZ / XLSX** work on any package, including one built by hand or relinked.

PART G

Technical reference

The methods in brief, the validation record, the limits, and what to check before the numbers go into a design. `DOCUMENTATION.md` in the repository gives every method in full.

31 Methods and equations

Step	Method
Fill	Priority flood (Barnes, Lehman and Mulla 2014), O(N log N), optional minimum gradient ε per step across filled flats.
D8	Steepest descent $S_k = (z_i - z_k) / d_k$, d_k = cell size or cell size·√2; ties within 1e-6 by the order S, W, N, E, SE, SW, NW, NE.
Flats	Barnes (2014): gradients away from high edges and towards low edges combined as 2·toward + (height − away), iterated to convergence for nested flats; or toward-lower by breadth-first distance to the outlet. Flats touching the grid edge or NoData drain to it.
Accumulation	Topological sweep of the receiver graph (in-degree queue), own cell excluded; cycles reported.
Strahler	On stream cells only: a source is 1; two or more equal upstream maxima increment the order.
Snapping	Nearest stream cell within the radius, ties by accumulation.
Longest flow path	Upstream distance propagated through the receiver tree with true diagonal weights; the path walked back from the most distant divide cell.
10–85 slope	$(z_{85} - z_{10}) / (L_{85} - L_{10})$, L_{10} = 0.10 L and L_{85} = 0.85 L from the outlet, z interpolated linearly along the path (since 0.9; up to 0.8.3 the points were measured from the divide).
Catchment slope	Horn 3 × 3 mean gradient over the catchment cells; relief ratio = relief / LFP length.
Polygons	Each catchment traced from its own cell mask (OGC-valid, identical to `gdal.Polygonize` followed by a union).
Crossings	D8 link (cell centre to receiver centre) × centreline segments; clusters by union-find over parallel runs and the merge distance.
USLE K	Williams (EPIC): $K = f_{csand}\, f_{cl\text{-}si}\, f_{orgc}\, f_{hisand}$, $f_{csand} = 0.2 + 0.3 \exp[-0.0256\,\mathrm{SAN}\,(1 - \mathrm{SIL}/100)]$; SI K = 0.1317 × US K.
Curve number	TR-55 Table 2-2 by land cover × HSG; $CN_I = 4.2\,CN / (10 - 0.058\,CN)$, $CN_{III} = 23\,CN / (10 + 0.13\,CN)$.
R from rainfall	Renard and Freimund (1994): $0.0483\,P^{1.61}$ ($P \le 850$ mm), $587.8 - 1.219\,P + 0.004105\,P^2$ above; Lo et al. (1985): $38.46 + 3.48\,P$.
LS, SPI, TWI, STI	$LS = (A_s/22.13)^m (\sin\beta / 0.0896)^{1.3}$; $SPI = A_s \tan\beta$; $TWI = \ln(A_s / \tan\beta)$; STI = LS with m = 0.6 on overland cells. tan β floored at 0.001 in the indices only.
Sediment	$SDR = 0.565\,A^{-0.125}$ (capped at 1); impact score 0.4 SPI + 0.3 RUSLE + 0.3 sediment on a 1–5 scale.

Step	Method
Tc	Kirpich 0.0195 $L^{0.77}$ $S^{-0.385}$; Kerby 1.44 $(L\ N)^{0.467}$ $S^{-0.235}$; SCS lag $\ell^{0.8}$ $(S + 1)^{0.7}$ / $(1900\ Y^{0.5})$; TR-55 sheet 0.007 $(n\ L)^{0.8}$ / $(P2^{0.5}\ s^{0.4})$, shallow 16.1345 $s^{0.5}$ ft/s, channel Manning; Bransby-Williams 14.6 L / $(A^{0.1}\ S^{0.2})$.
HAND	Height above the first stream cell on each cell's D8 path, by pointer jumping.
DEM error field	White noise convolved by FFT with $\exp(-2r^2/L^2)$ and scaled to unit variance, so the correlation is $\exp(-r^2/L^2)$; multiplied by σ.

32 Validation

Analytic and software tests

25 pure-Python suites with 676 checks run without QGIS: analytic planes and valleys with exact areas, lengths and slopes; trapezoidal channels with exact widths; the v0.8.3 reference implementations bit for bit; Barnes against a per-cell port of RichDEM on random surfaces with NoData; hand-computed curve numbers, K and Tc; the A14 erosion study's tables; and the package against its golden fixture. A smoke test runs every Processing tool inside QGIS (80 checks) and an end-to-end test runs the pipeline on a synthetic road site with known structure (25 checks), on QGIS 3.34.4 and 4.2.3. Bandit reports no findings.

Against a commercial reference toolset (Site A, steep)

Quantity, 16 road crossings, 30.9 m DEM	Result
Catchment area	median ratio 1.009; 15 of 16 within ±30 %
Spatial agreement (IoU)	median 0.84
Longest flow path length	median ratio 0.999; 15 of 16 within ±20 %
Flow direction, aligned grid	98.3 % identical; disagreement concentrated in filled depressions (24 % there, 1.2 % elsewhere)

Flat terrain (Site B, coastal, 34 % flat cells)

Flow direction agrees on 88 % of cells; the difference lies in the flats and is a difference of algorithm, not a defect. Fed the reference toolset's own flow direction, QEHT's accumulation reproduces its accumulation exactly on 99.31 % of 9 million cells, which isolates any difference to flat routing.

National benchmark

80 stratified random 15 km areas across Kenya (20 each flat, rolling, hilly and mountainous), each on ALOS AW3D30 and FABDEM, routed with QEHT and with TauDEM, RichDEM and two commercial packages. The two flat methods differ by more than TauDEM and RichDEM differ from each other in every stratum; Barnes agrees best with TauDEM. `BENCHMARK.md` has the design, tables and decision.

33 Known limitations

- **No tiling.** The whole grid is processed in memory. Clip to the catchments plus a buffer, or use the auto-clip; results inside the clip are unchanged.
- **Flat terrain.** On large flats the flat method decides the network and catchments switch between crossings under DEM error. QEHT flags this (flat check, uncertainty, divergence); it cannot resolve it. Survey or a better DEM can.
- **30 m DEMs.** Channels narrower than about three cells, road embankments and small culverts are not in the terrain. Channel sections often find the valley shoulder.
- **Proxies.** The WorldCover-based curve numbers, C factors and Tc retardance values are matches, not measurements. Replace them with local lookups where they exist.

- **Single-threaded.** The core is vectorised NumPy but uses one processor.
- **No hydraulics.** QEHT stops at the catchment and its inputs: no design discharge, culvert sizing or flood levels.

34 Quality-control checklist

1. The DEM is one product, projected in metres, resampled bilinearly; the NoData and resampling audits are clean (or the NoData override is set).
2. The stream threshold suits the smallest catchment you design for, and the snap threshold catches the tributaries the crossings sit on.
3. Every crossing is reviewed against imagery or survey: accepted, rejected, moved or added; `snap_dist_m` is small.
4. Flat-sensitive crossings (`flat_sensitive = 1`) and crossings that switch under DEM error are checked on a map or on site.
5. Proposed crossings, flat stretches and mapped rivers without a crossing have each been decided.
6. `acc_at_outlet_km2` equals `area_km2` for every full catchment.
7. The design report states the slope used (and its field), the Tc method and its validity flag, the curve-number source and condition, and which values are proxies or estimates.
8. The QEHT version, flat method and DEM SHA-256 from `qeht_run_metadata` are recorded with the design.

35 Troubleshooting

Symptom	Cause and remedy
"A projected, metric CRS is required"	The DEM is in degrees. Run Prepare DEM for hydrology.
Every catchment drains to the edge or one huge sink	Undeclared NoData. Set the NoData override to the value the audit reports.
Parallel stripes of streams; many flat-sensitive crossings	Nearest-neighbour resampling or whole-metre flats. Prepare the native DEM bilinearly; consider a minimum fill gradient of 0.0001.
A catchment far too large	The pour point snapped to the trunk stream. Lower the snap-to-stream threshold or the snap radius, or move the point onto the tributary.
A lake behind the road in the filled DEM	The embankment is in the DEM. Burn the crossings.
"No candidate is accepted or recommended"	Every candidate is rejected. Accept at least one, or give pour points.
Duplicate or empty IDs	The ID attribute has repeats or blanks, or two points snap to one cell. The message lists them.
Memory warning	Clip to the road's contributing area or to the catchments plus a buffer.
TR-55 Tc empty	Give P2.
No soil loss, only "terrain potential"	R, K or C is missing; give the missing factor.
KMZ without legend and title block	Matplotlib is not installed with this QGIS; the rest of the KMZ is complete.

APPENDIX A

Parameter reference

Every tool with its parameter names (for scripts and models), labels and defaults, generated from the plugin itself.

Workflow

Export package to KMZ / XLSX · `qeht:exportpackage`

Writes a design hydrology package as a KMZ for Google Earth and as an XLSX workbook, both in pure Python (no extra software).

Name	Label	Default
PACKAGE	Design hydrology package	—
TITLE	Title (KMZ document and title block)	—
RELIEF	KMZ: include the relief quicklook as a terrain overlay (if present)	True
KMZ	KMZ	output
XLSX	XLSX workbook	output

Run hydrology pipeline (one click) · `qeht:hydrologypipeline`

Runs the whole QEHT chain in one go and writes one output folder with a fixed layout: layers/ (crossings, catchments, flow paths, streams, candidates and the QA layers), rasters/ (filled DEM, flow direction, accumulation, streams, Strahler order, erosion), tables/ (the catchment characteristics table - one row per crossing, ready for a spreadsheet - and one CSV per layer), exports/ (KMZ for Google Earth and an XLSX workbook), quicklooks/ (PNG + world file + legend of the rasters, for viewing without a GIS), report/run_report.html and settings.json.

Name	Label	Default
SETTINGS	Re-run from settings (settings.json of an earlier run; optional)	—
DEM	DEM (raw, projected, metres)	—
PREPARE	Prepare the DEM first (merge tiles, reproject bilinear to the local UTM zone at the native cell size, audit)	False
DEM_TILES	DEM tiles to prepare (optional; else the DEM above)	—
AUTO_CLIP	Clip the DEM to the road's contributing area first (large DEMs; needs a road)	False
ROAD	Road alignment (centreline; optional)	—
CROSSINGS	Crossings: reviewed candidates or pour points (optional with a road)	—
START	Start chainage (m)	0.0
REVERSE	Reverse chainage direction	False
STAGE	Run	Full run

Name	Label	Default
STREAM_KM2	Stream threshold (km2)	1.0
FLAT_METHOD	Flat resolution	Barnes 2014 (recommended)
BURN	Burn crossings through embankments, then route again (needs a road)	False
NODATA_OVERRIDE	Treat this DEM value as NoData (blank = trust the file header)	—
DEM_SOURCE	DEM source for the record (e.g. FABDEM 30 m, bilinear to UTM 37S)	—
RUN_NAME	Run name (report title)	—
SOIL_POLYGONS	Soil map polygons (SOTWIS / SOTER, national map, any polygons)	—
SOIL_DB	SOTWIS / SOTER SQLite database (optional; full unit composition)	—
SOIL_CSV	Soil table CSV (optional; unit field + sand, silt, clay, oc %, [bulk, cfrag, drain])	—
SOIL_UNIT_FIELD	Soil unit field linking polygons / unit raster to the database or table	NEWSUID
SOIL_TOP	Soil depth from (cm)	0.0
SOIL_BOTTOM	Soil depth to (cm)	20.0
SOIL_F_SAND	Soil polygons: field with sand % (optional) (advanced)	—
SOIL_F_SILT	Soil polygons: field with silt % (optional) (advanced)	—
SOIL_F_CLAY	Soil polygons: field with clay % (optional) (advanced)	—
SOIL_F_OC	Soil polygons: field with organic carbon (optional) (advanced)	—
SOIL_F_BULK	Soil polygons: field with bulk density g/cm3 (optional) (advanced)	—
SOIL_F_CFRAG	Soil polygons: field with coarse fragments vol % (optional) (advanced)	—
SOIL_F_DRAIN	Soil polygons: field with FAO drainage class (optional) (advanced)	—
SOIL_F_HSG_FIELD	Soil polygons: field with hydrologic soil group (A, B, C, D, A/D ...) (optional) (advanced)	—
SOIL_F_K_FIELD	Soil polygons: field with USLE K, SI units (optional) (advanced)	—
SOIL_OC_UNITS	Soil polygons: organic carbon field units (advanced)	percent
SOIL_SOILGRIDS	SoilGrids folder (sand/silt/clay/soc/bdod/cfvo_<depth>cm_mean.tif; depth-weighted) (advanced)	—
SOIL_SAND_R	Sand raster (optional) (advanced)	—
SOIL_CLAY_R	Clay raster (optional) (advanced)	—

Name	Label	Default
SOIL_SILT_R	Silt raster (optional; else 100 - sand - clay) (optional) (advanced)	—
SOIL_OC_R	Organic carbon raster (optional) (advanced)	—
SOIL_BULK_R	Bulk density raster (optional) (advanced)	—
SOIL_CFRAG_R	Coarse fragments raster (optional) (advanced)	—
SOIL_RASTER_UNITS	Units of the soil rasters (advanced)	SoilGrids 2.0 (g/kg; soc dg/kg; bdod cg/cm3; cfvo per mille)
SOIL_UNIT_RASTER	Soil unit raster (codes joined to the CSV table, e.g. HWSD v2) (advanced)	—
SOIL_HSG_R	Hydrologic soil group raster (HYSOGs250m codes 1-4, 11-14) (advanced)	—
SOIL_K_R	USLE K raster, SI (overrides K computed from texture) (advanced)	—
LANDCOVER	Land cover (ESA WorldCover classes; optional, for CN and land-cover shares)	—
CN_CONDITION	Hydrologic condition for the TR-55 curve numbers	fair
CN_AMC	Antecedent moisture condition of the exported CN	II (average)
LANDCOVER_SCN	Land-cover scenario (same classes; optional, e.g. a planned development)	—
SCENARIO_NAME	Scenario name	—
CN_CSV	Curve number lookup CSV (class, A, B, C, D; replaces the TR-55 default) (advanced)	—
RC_CSV	Rational C lookup CSV (class, A, B, C, D or class, C; none ships) (advanced)	—
RAIN_ZONES	Rainfall zones (polygons; optional)	—
RAIN_ZONE_FIELD	Zone name field	—
RAIN_MAP	Mean annual rainfall raster, mm/yr (optional, e.g. a CHIRPS climatology)	—
RAIN_MAP_DATASET	Rainfall dataset name for the record (e.g. CHIRPS v2.0 1991-2020) (advanced)	—
RAIN_R_RELATION	RUSLE R from rainfall (an ESTIMATE; prefer an erosivity raster) (advanced)	none
RAIN_ZONE_QA	Warn when the dominant zone covers less than (%) (advanced)	80.0
MAPPED	Mapped waterways for the drainage check (lines; optional)	—
MAPPED_NAME_FIELD	Mapped waterways: name field	—
MAPPED_KM2	Mapped drainage: comparison threshold (km2; about 1 for OSM, 10 for HydroRIVERS)	1.0
TC_P2	2-yr 24-h rainfall P2 for the TR-55 time of concentration (mm; optional, no default)	—
EROSION	Erosion indices and RUSLE (LS-only when R, K or C is missing)	True
R_RASTER	R erosivity raster (e.g. GloREDa; optional)	—
R_VALUE	R single value (used when no raster)	—

Name	Label	Default
COVERAGE	Drainage coverage check along the road	True
QUICKLOOKS	Raster quicklooks (PNG + world file + legend, for viewing without a GIS)	True
UNC	DEM uncertainty at crossings (Monte Carlo; slow)	False
UNC_PRESET	DEM uncertainty: vertical error	Choose...
UNC_SIGMA	DEM uncertainty: custom sigma (m) (advanced)	—
UNC_CORR	DEM uncertainty: correlation length (m) (advanced)	90.0
UNC_N	DEM uncertainty: realisations (advanced)	50
UNC_SEED	DEM uncertainty: random seed (advanced)	2026
UNC_SENS	DEM uncertainty: correlation-length sensitivity (0.5x and 2x) for the three largest crossings (advanced)	True
EXPORTS	KMZ (Google Earth) and XLSX workbook in exports/	True
PACKAGE	Keep the design hydrology package (GeoPackage for design software)	False
LOAD	Load the main layers when finished	True
OUTPUT	Output folder	output

Run report from a design hydrology package · `qeht:runreport`

Writes a one-page HTML report for a design hydrology package: summary cards, a schematic plan (catchments, flow paths, road, crossings), the crossing schedule, the drainage coverage findings, inputs and provenance (DEM source and hash, conditioning, flat method, thresholds, soil / curve-number / rainfall / erosion sources with PROXY and ESTIMATE labels) and the curve number lookup.

Name	Label	Default
PACKAGE	Design hydrology package	—
TITLE	Report title	—
CSV	Also write the catchment characteristics table (CSV, next to the report)	True
OUTPUT	Run report	output

Terrain and drainage

Catchment and flow path characteristics · `qeht:catchmentcharacteristics`

Delineates a catchment and its longest flow path for each pour point, and writes the morphometry needed for a design flood calculation.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
DEM	Conditioned DEM	—
RAW_DEM	Raw DEM (for reported elevations)	—
FAC	Flow accumulation	—
POINTS	Pour points	—
SNAP	Snap radius (cells)	5

Name	Label	Default
SNAP_THRESHOLD	Snap-to-stream threshold (cells; 0 = max accumulation)	200.0
NESTED	Non-overlapping (local) catchments	False
ID_FIELD	ID attribute for outlet_uid (optional; blank = sequential)	—
ID_PREFIX	ID prefix (sequential IDs only; ignored with an ID attribute)	X
FLAT_CHECK	Flat-method check: area under both flat methods, flag differences	True
FLAT_TOL	Flat-method check tolerance (%)	10.0
SHEET_CAP	Sheet-flow cap within the overland part of the flow path (m)	100.0
CATCH_OUT	Catchments with characteristics	output
PATH_OUT	Longest flow paths with characteristics	output

D8 flow direction · `qeht:flowdirection`

Computes D8 flow direction by steepest descent, weighted by distance so that diagonal neighbours are compared over cellsize x sqrt(2).

Name	Label	Default
DEM	Conditioned DEM	—
RESOLVE_FLATS	Resolve flats	True
FLAT_METHOD	Flat resolution method	Barnes 2014 convergent (recommended)
OUTPUT	Flow direction (D8-coded)	output

Delineate catchments from pour points · `qeht:delineatecatchment`

Delineates the upstream contributing area for each pour point and writes both a labelled raster and dissolved polygons.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation (for snapping)	—
POINTS	Pour points	—
SNAP	Snap radius (cells)	5
SNAP_THRESHOLD	Snap-to-stream threshold (cells; 0 = snap to max accumulation)	200.0
ID_FIELD	ID attribute for outlet_uid (optional; blank = sequential)	—
ID_PREFIX	ID prefix (sequential IDs only; ignored with an ID attribute)	X
RASTER_OUT	Catchment raster	output
POLY_OUT	Catchment polygons	output

Fill depressions · `qeht:filldepressions`

Fills closed depressions in a DEM using the priority-flood algorithm (Barnes et al.

Name	Label	Default
DEM	Input DEM	—
MIN_SLOPE	Minimum slope across filled surfaces (m/m)	0.0
NODATA_OVERRIDE	Treat this value as NoData (leave blank to trust the header)	—

Name	Label	Default
OUTPUT	Conditioned DEM	output
DEPTH	Fill depth (QA)	output

Flow accumulation · `qeht:flowaccumulation`

Accumulates upstream contribution over a D8 flow direction raster by topological traversal - O(N), no recursion.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
QUANTITY	Accumulated quantity	Cell count (standard D8 convention)
WEIGHT	Weight raster	—
OUTPUT	Flow accumulation	output

Longest flow path · `qeht:longestflowpath`

Traces the longest flow path from the catchment divide to each pour point, and reports its length, elevation drop and average slope.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
DEM	Conditioned DEM (for slope)	—
FAC	Flow accumulation (for snapping)	—
POINTS	Pour points	—
SNAP	Snap radius (cells)	5
SNAP_THRESHOLD	Snap-to-stream threshold (cells; 0 = max accumulation)	200.0
NESTED	Non-overlapping (local) catchments	False
ID_FIELD	ID attribute for outlet_uid (optional; blank = sequential)	—
ID_PREFIX	ID prefix (sequential IDs only; ignored with an ID attribute)	X
OUTPUT	Longest flow paths	output

Prepare DEM for hydrology · `qeht:preparedem`

Merges DEM tiles, clips them, reprojects them to a projected metric CRS and checks the result, all in-process with GDAL.

Name	Label	Default
DEMS	DEM raster(s) (tiles of one product)	—
CLIP	Clip to polygons (optional)	—
EXTENT	Clip to extent (optional; used when no clip polygons)	—
BUFFER	Clip buffer (m)	2000.0
CRS_MODE	Target CRS	Auto: WGS 84 / UTM zone of the extent centre (UPS near the poles)
TARGET_CRS	Target CRS (when chosen above)	—
CELL	Cell size (m; blank = native, converted to metres)	—
RESAMPLING	Resampling	Bilinear (recommended)

Name	Label	Default
NODATA	Treat this input value as NoData (optional)	—
SOURCE	DEM source for the record (e.g. FABDEM v1.2)	—
OUTPUT	Prepared DEM	output
LOG	Preparation log	output

Stream network and Strahler order · `qeht:streamnetwork`

Extracts a channel network by thresholding flow accumulation, then assigns Strahler order to the extracted network.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation (cell count)	—
MODE	Threshold type	Accumulation threshold (cells)
THRESHOLD	Threshold value	1000.0
STREAMS	Stream mask	output
ORDER	Strahler order	output

Stream network to polylines · `qeht:streamlines`

Extracts the channel network at a flow accumulation threshold and traces it into polyline REACHES - one feature per channel segment between sources, junctions and the outlet.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation (cell count)	—
DEM	Conditioned DEM (for slope)	—
MODE	Threshold type	Accumulation threshold (cells)
THRESHOLD	Threshold value	1000.0
OUTPUT	Stream reaches	output

Road drainage

Alignment ground profile · `qeht:alignmentprofile`

Samples the ground along a road centreline every step (default 10 m) of chainage: DEM ground (bilinear), filled-DEM level and the ponding depth the fill removed, contributing area, the stream crossings and their Strahler order, and the longitudinal ground slope.

Name	Label	Default
RAW_DEM	Raw DEM	—
ROAD	Road centreline	—
FILLED	Filled DEM (optional; ponding depth)	—
FDR	Flow direction (D8-coded; optional, for stream crossings)	—

Name	Label	Default
FAC	Flow accumulation (optional; contributing area)	—
ORDER	Stream order raster (optional)	—
THRESHOLD	Stream threshold (cells)	200.0
START	Start chainage (m)	0.0
REVERSE	Reverse chainage direction	False
STEP	Station spacing (m)	10.0
NAME	Alignment name (written to every station)	—
PROFILE	Alignment profile stations	output
CSV	Profile table (CSV)	output
CHART	Profile chart (PNG)	output

Burn crossings through embankments · `qeht:burncrossings`

A DSM or survey DEM shows a road embankment as a dam; the culvert under it is invisible.

Name	Label	Default
DEM	Raw DEM	—
CROSSINGS	Crossings (candidate layer or points)	—
ROAD	Road alignment (needed if crossings carry no road_azimuth_deg)	—
ALL	Burn every point (ignore candidate status)	False
HALF	Breach half-length either side of the centreline (m)	30.0
SEARCH	Search radius for the breach ends (cells)	2
OUTPUT	Burned DEM	output
LOG	Breach log	output

Clip DEM to the road's contributing area · `qeht:autoclip`

Cuts a large DEM down to the area that drains to the road, so the rest of QEHT fits in memory, without cutting catchments by hand.

Name	Label	Default
DEM	DEM (projected, metres)	—
ROAD	Road alignment	—
BUFFER	Road buffer (m)	200.0
MARGIN	Margin around the contributing area (m)	1000.0
MAX_MCELLS	Coarse grid at most (million cells) (advanced)	4.0
MODE	Clip to	Bounding box (recommended)
OUTPUT	Clipped DEM	output

DEM uncertainty at crossings · `qeht:demuncertainty`

Turns each crossing's catchment area, flow-path length, 10-85 slope and Kirpich Tc into a range under DEM error (Monte Carlo).

Name	Label	Default
DEM	Raw DEM (projected, metres)	—
CROSSINGS	Crossings (candidates, a package's crossings or pour points)	—
ID_FIELD	ID field (e.g. outlet_uid)	—
THRESHOLD	Stream threshold (cells)	200.0
SNAP	Snap radius (cells)	5
UNC_PRESET	DEM uncertainty: vertical error	Choose...
UNC_SIGMA	DEM uncertainty: custom sigma (m) (advanced)	—
UNC_CORR	DEM uncertainty: correlation length (m) (advanced)	90.0
UNC_N	DEM uncertainty: realisations (advanced)	50
UNC_SEED	DEM uncertainty: random seed (advanced)	2026
UNC_SENS	DEM uncertainty: correlation-length sensitivity (0.5x and 2x) for the three largest crossings (advanced)	True
OUTPUT	Crossings with DEM uncertainty	output
LOG	Settings and correlation-length sensitivity (JSON)	output

Drainage coverage check along a road · `qeht:drainagecoverage`

Checks a road alignment for drainage that has no crossing, from the DEM alone.

Name	Label	Default
RAW_DEM	Raw DEM	—
FILLED	Filled DEM (optional; filled from the raw DEM if not given)	—
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation	—
ROAD	Road centreline	—
CROSSINGS	Existing crossings (pour points or crossing candidates; optional)	—
START	Start chainage (m)	0.0
REVERSE	Reverse chainage direction	False
THRESHOLD	Stream threshold (cells)	200.0
MIN_AREA	Smallest stream area to need a crossing (km2; 0 = stream threshold)	0.0
SEARCH	An existing crossing within this chainage covers a stream (m)	50.0
MERGE	Merge stream stations within (m)	30.0
STEP	Profile station spacing (m)	10.0
SAG_SMOOTH	Sags: smoothing window (m)	30.0
SAG_DEPTH	Sags: minimum depth (m)	0.3
FLAT_SLOPE	Flat stretches: slope below (%)	0.5
FLAT_CROSSFALL	Flat stretches: cross-fall measured over (m)	100.0
FLAT_MIN_LEN	Flat stretches: minimum length (m)	300.0
COVERAGE_OUT	Coverage findings	output

Name	Label	Default
SAGS_OUT	Sag points	output
FLATS_OUT	Flat stretches	output

Road crossing candidates · `qeht:crossingcandidates`

Finds every place the drainage network crosses a road alignment and proposes one crossing per drainage line, without deleting anything.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation	—
STREAMS	Stream raster (optional; blank = accumulation threshold)	—
THRESHOLD	Stream threshold (cells) when no stream raster is given	200.0
ORDER	Stream order raster (optional)	—
ROAD	Road alignment (centreline)	—
START	Start chainage (m)	0.0
REVERSE	Reverse chainage direction	False
MIN_AREA	Minimum contributing area (km2)	0.01
HALFWIDTH	Corridor half-width for parallel flow (m)	30.0
MIN_PARALLEL	Minimum parallel length (m)	100.0
MERGE	Cluster merge distance along chainage (m; 0 = off)	0.0
PREFIX	Candidate ID prefix	C
CANDIDATES	Crossing candidates	output
PARALLEL	Parallel reaches (side-drain hints)	output

Soils and erosion

Erosion indices and RUSLE soil loss · `qeht:erosionindices`

Terrain erosion indices, optional RUSLE soil loss and erosion severity classes, written to one output folder that the corridor sampler and the design hydrology package read.

Name	Label	Default
RAW_DEM	Raw (unfilled) DEM	—
FAC	Flow accumulation (cells, from the conditioned DEM)	—
LS_METHOD	LS factor	Moore & Burch (1986), m = 0.4 (default)
CHANNEL_CELLS	Channel threshold (upslope cells) for the channel/hillslope split	1000.0
R_RASTER	R erosivity raster	—
R_VALUE	R erosivity single value (used when no raster)	—
K_RASTER	K erodibility (SI) raster	—
K_VALUE	K erodibility (SI) single value (used when no raster)	—

Name	Label	Default
C_RASTER	C cover raster	—
C_VALUE	C cover single value (used when no raster)	—
R_MAP	Mean annual rainfall raster, mm/yr (R ESTIMATE when no R is given) (advanced)	—
R_RELATION	R-P relation for the estimate (advanced)	Renard & Freimund 1994
SOIL_POLYGONS	Soil map polygons (SOTWIS / SOTER, national map, any polygons)	—
SOIL_DB	SOTWIS / SOTER SQLite database (optional; full unit composition)	—
SOIL_CSV	Soil table CSV (optional; unit field + sand, silt, clay, oc %, [bulk, cfrag, drain])	—
SOIL_UNIT_FIELD	Soil unit field linking polygons / unit raster to the database or table	NEWSUID
SOIL_TOP	Soil depth from (cm)	0.0
SOIL_BOTTOM	Soil depth to (cm)	20.0
SOIL_F_SAND	Soil polygons: field with sand % (optional) (advanced)	—
SOIL_F_SILT	Soil polygons: field with silt % (optional) (advanced)	—
SOIL_F_CLAY	Soil polygons: field with clay % (optional) (advanced)	—
SOIL_F_OC	Soil polygons: field with organic carbon (optional) (advanced)	—
SOIL_F_BULK	Soil polygons: field with bulk density g/cm3 (optional) (advanced)	—
SOIL_F_CFRAG	Soil polygons: field with coarse fragments vol % (optional) (advanced)	—
SOIL_F_DRAIN	Soil polygons: field with FAO drainage class (optional) (advanced)	—
SOIL_F_HSG_FIELD	Soil polygons: field with hydrologic soil group (A, B, C, D, A/D ...) (optional) (advanced)	—
SOIL_F_K_FIELD	Soil polygons: field with USLE K, SI units (optional) (advanced)	—
SOIL_OC_UNITS	Soil polygons: organic carbon field units (advanced)	percent
SOIL_SOILGRIDS	SoilGrids folder (sand/silt/clay/soc/bdod/cfvo_<depth>cm_mean.tif; depth-weighted) (advanced)	—
SOIL_SAND_R	Sand raster (optional) (advanced)	—
SOIL_CLAY_R	Clay raster (optional) (advanced)	—
SOIL_SILT_R	Silt raster (optional; else 100 - sand - clay) (optional) (advanced)	—
SOIL_OC_R	Organic carbon raster (optional) (advanced)	—
SOIL_BULK_R	Bulk density raster (optional) (advanced)	—

Name	Label	Default
SOIL_CFRAG_R	Coarse fragments raster (optional) (advanced)	—
SOIL_RASTER_UNITS	Units of the soil rasters (advanced)	SoilGrids 2.0 (g/kg; soc dg/kg; bdod cg/cm3; cfvo per mille)
SOIL_UNIT_RASTER	Soil unit raster (codes joined to the CSV table, e.g. HWSD v2) (advanced)	—
SOIL_HSG_R	Hydrologic soil group raster (HYSOGs250m codes 1-4, 11-14) (advanced)	—
SOIL_K_R	USLE K raster, SI (overrides K computed from texture) (advanced)	—
WORLDCOVER	ESA WorldCover 2021 classes (for C, and P if chosen)	—
P_RASTER	P support raster	—
P_VALUE	P single value	1.0
P_FROM_WORLDCOVER	P from WorldCover (A14 2025 lookup) instead of the value	False
SPI_SCHEME	SPI classes	Percentiles 50/75/90/97 of this extent (relative, D7)
SPI_BREAKS	Custom ln(SPI) breaks (comma-separated; overrides the scheme)	—
RUSLE_SCHEME	RUSLE classes	t/ha/yr 5 / 12 / 25 / 50: Very low ... Very high (D7)
RUSLE_BREAKS	Custom RUSLE breaks t/ha/yr (4 values; overrides the scheme)	—
MATRIX	Combination matrix, 25 values row by row (SPI score x RUSLE/LS score; blank = higher of the two)	—
BULK_DENSITY	Bulk density for sediment volume when no soil data (kg/m3)	1400.0
WEIGHTS	Crossing impact weights SPI, RUSLE, sediment	0.4,0.3,0.3
STI	Also write the overland sediment transport capacity index (STI, channels masked)	True
STI_CAP	STI: cap on specific catchment area (m; validity ~100 m slope length) (advanced)	100.0
DEP_NEAR	Deposition indicator: near reach upstream of a crossing (m) (advanced)	100.0
DEP_FAR	Deposition indicator: far reach ends this far upstream (m) (advanced)	500.0
DEP_BREAKS	Deposition indicator breaks (ratio below = deposition-prone, above = scour-prone) (advanced)	0.7,1.3
OUTPUT	Erosion output folder	output

Sample erosion along alignment · `qeht:erosioncorridor`

Samples the erosion folder written by 'Erosion indices and RUSLE soil loss' along a road centreline.

Name	Label	Default
EROSION	Erosion output folder	—
ROAD	Road centreline	—

Name	Label	Default
START	Chainage at the start of the line (m)	0.0
REVERSE	Reverse the chainage direction	False
STEP	Station spacing (m)	10.0
HALF_WIDTH	Buffer half-width each side (m)	50.0
OFFSET_STEP	Sampling interval across the road (m)	10.0
FDR	Flow direction (D8-coded; optional, for the side-drain siltation indicator)	—
RAW_DEM	Raw DEM (optional; longitudinal slope for the siltation flag)	—
ROAD_HALF	Side drains: road strip half-width (m) (advanced)	5.0
DRAIN_SLOPE	Side drains: flag below this longitudinal slope (%) (advanced)	1.0
STATIONS	Erosion stations	output
REACHES	Erosion reaches	output
CHART	Chainage profile chart	output

Soil, rainfall and runoff parameters for catchments · `qeht:soilparameters`

Adds a soil block to catchment polygons (any polygon layer - e.g.

Name	Label	Default
CATCHMENTS	Catchment polygons	—
REF	Reference grid (the DEM or flow-direction raster used for the catchments)	—
SOIL_POLYGONS	Soil map polygons (SOTWIS / SOTER, national map, any polygons)	—
SOIL_DB	SOTWIS / SOTER SQLite database (optional; full unit composition)	—
SOIL_CSV	Soil table CSV (optional; unit field + sand, silt, clay, oc %, [bulk, cfrag, drain])	—
SOIL_UNIT_FIELD	Soil unit field linking polygons / unit raster to the database or table	NEWSUID
SOIL_TOP	Soil depth from (cm)	0.0
SOIL_BOTTOM	Soil depth to (cm)	20.0
SOIL_F_SAND	Soil polygons: field with sand % (optional) (advanced)	—
SOIL_F_SILT	Soil polygons: field with silt % (optional) (advanced)	—
SOIL_F_CLAY	Soil polygons: field with clay % (optional) (advanced)	—
SOIL_F_OC	Soil polygons: field with organic carbon (optional) (advanced)	—
SOIL_F_BULK	Soil polygons: field with bulk density g/cm3 (optional) (advanced)	—
SOIL_F_CFRAG	Soil polygons: field with coarse fragments vol % (optional) (advanced)	—

Name	Label	Default
SOIL_F_DRAIN	Soil polygons: field with FAO drainage class (optional) (advanced)	—
SOIL_F_HSG_FIELD	Soil polygons: field with hydrologic soil group (A, B, C, D, A/D ...) (optional) (advanced)	—
SOIL_F_K_FIELD	Soil polygons: field with USLE K, SI units (optional) (advanced)	—
SOIL_OC_UNITS	Soil polygons: organic carbon field units (advanced)	percent
SOIL_SOILGRIDS	SoilGrids folder (sand/silt/clay/soc/bdod/cfvo_<depth>cm_mean.tif; depth-weighted) (advanced)	—
SOIL_SAND_R	Sand raster (optional) (advanced)	—
SOIL_CLAY_R	Clay raster (optional) (advanced)	—
SOIL_SILT_R	Silt raster (optional; else 100 - sand - clay) (optional) (advanced)	—
SOIL_OC_R	Organic carbon raster (optional) (advanced)	—
SOIL_BULK_R	Bulk density raster (optional) (advanced)	—
SOIL_CFRAG_R	Coarse fragments raster (optional) (advanced)	—
SOIL_RASTER_UNITS	Units of the soil rasters (advanced)	SoilGrids 2.0 (g/kg; soc dg/kg; bdod cg/cm3; cfvo per mille)
SOIL_UNIT_RASTER	Soil unit raster (codes joined to the CSV table, e.g. HWSD v2) (advanced)	—
SOIL_HSG_R	Hydrologic soil group raster (HYSOGs250m codes 1-4, 11-14) (advanced)	—
SOIL_K_R	USLE K raster, SI (overrides K computed from texture) (advanced)	—
LANDCOVER	Land cover (ESA WorldCover classes; optional, for CN and land-cover shares)	—
CN_CONDITION	Hydrologic condition for the TR-55 curve numbers	fair
CN_AMC	Antecedent moisture condition of the exported CN	II (average)
LANDCOVER_SCN	Land-cover scenario (same classes; optional, e.g. a planned development)	—
SCENARIO_NAME	Scenario name	—
CN_CSV	Curve number lookup CSV (class, A, B, C, D; replaces the TR-55 default) (advanced)	—
RC_CSV	Rational C lookup CSV (class, A, B, C, D or class, C; none ships) (advanced)	—
RAIN_ZONES	Rainfall zones (polygons; optional)	—
RAIN_ZONE_FIELD	Zone name field	—
RAIN_MAP	Mean annual rainfall raster, mm/yr (optional, e.g. a CHIRPS climatology)	—
RAIN_MAP_DATASET	Rainfall dataset name for the record (e.g. CHIRPS v2.0 1991-2020) (advanced)	—
RAIN_R_RELATION	RUSLE R from rainfall (an ESTIMATE; prefer an erosivity raster) (advanced)	none

Name	Label	Default
RAIN_ZONE_QA	Warn when the dominant zone covers less than (%) (advanced)	80.0
OUTPUT	Catchments with soil parameters	output
K_RASTER	USLE K raster (optional)	output
UNIT_RASTER	Soil unit index raster (optional)	output

Interoperability

Build design hydrology package · `qeht:buildheasexchange`

One self-describing GeoPackage of crossings, catchments and flow paths linked by outlet_uid, for spreadsheets, reports or design software.

Name	Label	Default
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation	—
RAW_DEM	Raw DEM (for reported elevations and slopes)	—
ORDER	Stream order raster (optional, for crossing attributes)	—
POINTS	Crossings / pour points	—
ROAD	Road alignment (optional; chainage for hand-placed points)	—
START	Start chainage (m)	0.0
REVERSE	Reverse chainage direction	False
ID_FIELD	ID attribute (optional; blank = sequential IDs)	—
ID_PREFIX	ID prefix (sequential IDs only; ignored with an ID attribute)	X
ID_ORDER	Sequential numbering order	Automatic: along the chainage when known, else downstream first
SNAP	Snap radius (cells)	5
SNAP_THRESHOLD	Snap-to-stream threshold (cells; 0 = max accumulation)	200.0
LOCAL	Non-overlapping (local) catchments	False
DEM_SOURCE	DEM source (recorded in metadata, e.g. 'ALOS 30 m')	—
CONDITIONING	Conditioning used (recorded, e.g. 'fill, min_slope 0')	—
FLAT_METHOD	Flat resolution used for flow direction (recorded, e.g. 'barnes')	—
COVERAGE	Coverage check with a road alignment: missing crossings, sags and flat stretches; delineate proposed crossings	True
COV_MIN_AREA	Coverage: smallest stream area to need a crossing (km2; 0 = stream threshold) (advanced)	0.0
COV_SEARCH	Coverage: an existing crossing within this chainage covers a stream (m) (advanced)	50.0

Name	Label	Default
COV_MERGE	Coverage: merge stream stations within (m) (advanced)	30.0
COV_SMALL	Coverage: flag existing crossings smaller than (km2) (advanced)	0.01
SAG_SMOOTH	Sags: smoothing window along the profile (m) (advanced)	30.0
SAG_DEPTH	Sags: minimum depth (m) (advanced)	0.3
SAG_MIN_AREA	Sags: area that makes a sag a proposed crossing (km2) (advanced)	0.05
FLAT_SLOPE	Flat stretches: slope below (%) (advanced)	0.5
FLAT_CROSSFALL	Flat stretches: cross-fall measured over (m) (advanced)	100.0
FLAT_MIN_LEN	Flat stretches: minimum length (m) (advanced)	300.0
PROPOSED_PREFIX	Prefix for proposed crossings (advanced)	P
CH_SLOPE_DIST	Approach / exit channel length for the crossing channel slopes (m)	200.0
SHEET_CAP	Sheet-flow cap within the overland part of the flow path (m)	100.0
FILLED	Filled DEM (optional; ponding depth in the alignment profile)	—
QUICKLOOKS	Raster quicklooks folder (PNG + world file + legend; optional)	output
QL_MAX_PX	Quicklook size, long side (pixels) (advanced)	4096
FP_MIN_AREA	Floodplain width indicator: crossings of at least (km2) (advanced)	10.0
FP_HAND	Floodplain width indicator: also by height above nearest drainage (HAND) (advanced)	True
XS	Channel cross-section downstream of each crossing (indicative) (advanced)	True
XS_DIST	Channel section: distance downstream of the outlet (m) (advanced)	30.0
XS_HALF	Channel section: half width of the transect (m) (advanced)	150.0
XS_BANK_SLOPE	Channel section: bank top where the side slope falls below (m/m) (advanced)	0.05
PROFILE_STEP	Alignment profile station spacing (m)	10.0
SOIL_POLYGONS	Soil map polygons (SOTWIS / SOTER, national map, any polygons)	—
SOIL_DB	SOTWIS / SOTER SQLite database (optional; full unit composition)	—
SOIL_CSV	Soil table CSV (optional; unit field + sand, silt, clay, oc %, [bulk, cfrag, drain])	—
SOIL_UNIT_FIELD	Soil unit field linking polygons / unit raster to the database or table	NEWSUID
SOIL_TOP	Soil depth from (cm)	0.0

Name	Label	Default
SOIL_BOTTOM	Soil depth to (cm)	20.0
SOIL_F_SAND	Soil polygons: field with sand % (optional) (advanced)	—
SOIL_F_SILT	Soil polygons: field with silt % (optional) (advanced)	—
SOIL_F_CLAY	Soil polygons: field with clay % (optional) (advanced)	—
SOIL_F_OC	Soil polygons: field with organic carbon (optional) (advanced)	—
SOIL_F_BULK	Soil polygons: field with bulk density g/cm3 (optional) (advanced)	—
SOIL_F_CFRAG	Soil polygons: field with coarse fragments vol % (optional) (advanced)	—
SOIL_F_DRAIN	Soil polygons: field with FAO drainage class (optional) (advanced)	—
SOIL_F_HSG_FIELD	Soil polygons: field with hydrologic soil group (A, B, C, D, A/D ...) (optional) (advanced)	—
SOIL_F_K_FIELD	Soil polygons: field with USLE K, SI units (optional) (advanced)	—
SOIL_OC_UNITS	Soil polygons: organic carbon field units (advanced)	percent
SOIL_SOILGRIDS	SoilGrids folder (sand/silt/clay/soc/bdod/cfvo_<depth>cm_mean.tif; depth-weighted) (advanced)	—
SOIL_SAND_R	Sand raster (optional) (advanced)	—
SOIL_CLAY_R	Clay raster (optional) (advanced)	—
SOIL_SILT_R	Silt raster (optional; else 100 - sand - clay) (optional) (advanced)	—
SOIL_OC_R	Organic carbon raster (optional) (advanced)	—
SOIL_BULK_R	Bulk density raster (optional) (advanced)	—
SOIL_CFRAG_R	Coarse fragments raster (optional) (advanced)	—
SOIL_RASTER_UNITS	Units of the soil rasters (advanced)	SoilGrids 2.0 (g/kg; soc dg/kg; bdod cg/cm3; cfvo per mille)
SOIL_UNIT_RASTER	Soil unit raster (codes joined to the CSV table, e.g. HWSD v2) (advanced)	—
SOIL_HSG_R	Hydrologic soil group raster (HYSOGs250m codes 1-4, 11-14) (advanced)	—
SOIL_K_R	USLE K raster, SI (overrides K computed from texture) (advanced)	—
LANDCOVER	Land cover (ESA WorldCover classes; optional, for CN and land-cover shares)	—
CN_CONDITION	Hydrologic condition for the TR-55 curve numbers	fair
CN_AMC	Antecedent moisture condition of the exported CN	II (average)
LANDCOVER_SCN	Land-cover scenario (same classes; optional, e.g. a planned development)	—

Name	Label	Default
SCENARIO_NAME	Scenario name	—
CN_CSV	Curve number lookup CSV (class, A, B, C, D; replaces the TR-55 default) (advanced)	—
RC_CSV	Rational C lookup CSV (class, A, B, C, D or class, C; none ships) (advanced)	—
RAIN_ZONES	Rainfall zones (polygons; optional)	—
RAIN_ZONE_FIELD	Zone name field	—
RAIN_MAP	Mean annual rainfall raster, mm/yr (optional, e.g. a CHIRPS climatology)	—
RAIN_MAP_DATASET	Rainfall dataset name for the record (e.g. CHIRPS v2.0 1991-2020) (advanced)	—
RAIN_R_RELATION	RUSLE R from rainfall (an ESTIMATE; prefer an erosivity raster) (advanced)	none
RAIN_ZONE_QA	Warn when the dominant zone covers less than (%) (advanced)	80.0
BURN_LOG	Breach log from 'Burn crossings through embankments' (optional; recorded in the package)	—
EROSION	Erosion output folder (optional; adds the ero_* erosion block)	—
MAPPED	Mapped waterways for the drainage check (lines, e.g. OSM or HydroRIVERS; optional)	—
MAPPED_NAME_FIELD	Mapped waterways: name field	—
MAPPED_SOURCE	Mapped waterways: source for the record (e.g. OSM 2026-09) (advanced)	—
MAPPED_KM2	Mapped drainage: comparison threshold (km2; about 1 for OSM, 10 for HydroRIVERS) (advanced)	1.0
MAPPED_TOL	Mapped drainage: tolerance (m) (advanced)	60.0
MAPPED_MIN_CELLS	Mapped drainage: shortest divergence reach (cells) (advanced)	10.0
MAPPED_FAR	Mapped drainage: no comparison beyond (m from the outlet) (advanced)	1000.0
CSTI_HALF	Side-drain siltation: corridor half-width (m) (advanced)	50.0
CSTI_OFFSET	Side-drain siltation: sampling interval across the road (m) (advanced)	10.0
CSTI_ROAD_HALF	Side-drain siltation: road strip half-width (m) (advanced)	5.0
CSTI_DRAIN_SLOPE	Side-drain siltation: flag below this longitudinal slope (%) (advanced)	1.0
UNC	DEM uncertainty at crossings (Monte Carlo; slow)	False
UNC_PRESET	DEM uncertainty: vertical error	Choose...
UNC_SIGMA	DEM uncertainty: custom sigma (m) (advanced)	—
UNC_CORR	DEM uncertainty: correlation length (m) (advanced)	90.0
UNC_N	DEM uncertainty: realisations (advanced)	50

Name	Label	Default
UNC_SEED	DEM uncertainty: random seed (advanced)	2026
UNC_SENS	DEM uncertainty: correlation-length sensitivity (0.5x and 2x) for the three largest crossings (advanced)	True
TC_P2	Time of concentration: 2-yr 24-h rainfall P2 for TR-55 (mm; no default)	—
TC_P2_RASTER	Time of concentration: P2 raster (mm; mean along the flow path; optional)	—
TC	Time of concentration by five methods (advanced)	True
TC_CHANNEL_N	Time of concentration: channel Manning n (TR-55 channel part) (advanced)	0.035
TC_SHEET_N_CSV	Time of concentration: sheet-flow n lookup CSV (class, value; replaces the PROXY) (advanced)	—
TC_KERBY_N_CSV	Time of concentration: Kerby N lookup CSV (class, value; replaces the PROXY) (advanced)	—
CSV	Also write one CSV per layer (for spreadsheets)	False
OUTPUT	Exchange GeoPackage	output

Renumber and relink exchange package · `qeht:renumberrelink`

Proposed crossings (from the coverage check): set status to 'existing' (or 'adopted') to adopt one - it then gets the package prefix - delete it to drop it, or leave it 'proposed' to keep it numbered with the proposed prefix (P001...).

Name	Label	Default
PACKAGE	Edited exchange package	—
FDR	Flow direction (D8-coded)	—
FAC	Flow accumulation	—
RAW_DEM	Raw DEM (for reported elevations and slopes)	—
ORDER	Stream order raster (optional)	—
ROAD	Road alignment (optional; chainage of moved/added points)	—
START	Start chainage (m)	0.0
REVERSE	Reverse chainage direction	False
PREFIX	ID prefix (blank = keep the package's prefix)	—
PROPOSED_PREFIX	Prefix for crossings still proposed	P
SNAP	Snap radius for moved/added points (cells)	5
SNAP_THRESHOLD	Snap-to-stream threshold (cells)	200.0
LOCAL	Non-overlapping (local) catchments	False
OUTPUT	Relinked exchange package	output

APPENDIX B

Field dictionary

The fields of the design hydrology package, generated from its `qeht_field_dictionary` table, which every package carries.

`crossings` (104 fields)

Field	Unit	Meaning
outlet_uid	-	stable crossing/outlet identifier; identical on the crossing, its catchment and its longest flow path
crossing_id	-	crossing identifier (default = outlet_uid; editable)
outlet_id	-	pour-point feature id in the input layer - NOT stable, kept for backward compatibility only
source_id	-	value of the user-selected ID attribute, if any
input_x	m	pour point as supplied (DEM CRS)
input_y	m	pour point as supplied (DEM CRS)
outlet_x	m	snapped outlet (cell centre, DEM CRS)
outlet_y	m	snapped outlet (cell centre, DEM CRS)
snap_dist_m	m	distance moved by snapping
acc_at_outlet_km2	km2	contributing area at the snapped outlet read from the accumulation raster (equals area_km2 for full catchments)
stream_order	-	Strahler order at the outlet (if a stream-order raster was supplied)
chainage_m	m	position along the road alignment (from the crossing-candidate tool; empty for hand-placed pour points)
link_method	-	how this feature was linked to its crossing
link_confidence	0-1	confidence of the link
link_note	-	free-text note on the link
status	-	existing = in the design input; proposed = found by the coverage check and delineated in this run, not yet adopted
proposed_reason	-	uncovered_stream / sag_point (proposed crossings only)
nearest_uid	-	nearest existing crossing (proposed crossings only)
nearest_m	m	chainage distance to nearest_uid
ero_lnspi_max3x3	ln(m)	highest ln(SPI) in the 3×3 cells at the outlet
ero_lnspi_app_p50	ln(m)	median ln(SPI) on the approach channel
ero_lnspi_app_p90	ln(m)	90th percentile ln(SPI) on the approach channel
ero_slope_pct	%	local slope, 3×3 mean
ero_ls_local	-	local LS, 3×3 mean

Field	Unit	Meaning
ero_twi_local	ln(m)	local TWI, 3×3 mean
ero_spi_class	-	SPI class of ero_lnspi_max3×3
ero_spi_score	1-5	severity score of ero_spi_class
ero_worst_score_3x3	1-5	worst combined severity score in the 3×3 window
ero_composite	1-5	hydrodynamic impact score
ero_impact	-	impact level
ero_mitigation	-	indicative mitigation for the impact level
ero_sti_local	-	overland sediment transport capacity index, 3×3 mean
ero_spi_app_near	m	median SPI on the main channel 0 - near m upstream
ero_spi_app_far	m	median SPI on the main channel near - far m upstream
ero_dep_ratio	-	ero_spi_app_near / ero_spi_app_far
ero_dep_flag	-	deposition-prone / neutral / scour-prone
ero_slope_app_near_pct	%	channel slope of the near reach
ero_slope_app_far_pct	%	channel slope of the far reach
ero_dep_note	-	why no ratio was computed (e.g. channel too short)
ch_slope_us	m/m	approach channel slope upstream of the crossing
ch_slope_ds	m/m	exit channel slope downstream of the crossing
ch_len_us_m	m	length actually used upstream (shorter near the divide)
ch_len_ds_m	m	length actually used downstream (shorter at the grid edge)
ch_slope_dist_m	m	requested approach / exit length
fp_w_0p5_m	m	width along the road with ground below bed + 0.5 m
fp_w_1p0_m	m	width along the road below bed + 1.0 m
fp_w_2p0_m	m	width along the road below bed + 2.0 m
fp_ch_from_m	m	start chainage of the bed + 1.0 m extent
fp_ch_to_m	m	end chainage of the bed + 1.0 m extent
fp_z_bed_m	m	lowest ground on the profile within 50 m of the crossing
fp_hand_w_0p5_m	m	width along the road with HAND below 0.5 m
fp_hand_w_1p0_m	m	width with HAND below 1.0 m
fp_hand_w_2p0_m	m	width with HAND below 2.0 m
fp_method	-	profile or profile+HAND (empty: crossing below the area limit or no road)
fp_note	-	truncation at the end of the profile or NoData (width is a lower bound)
xs_dist_m	m	distance downstream of the outlet to the section
xs_bed_m	m	bed level at the section
xs_bed_offset_m	m	offset of the bed low point from the section centre
xs_bankfull_w_m	m	bank-full width (break of slope)
xs_bankfull_d_m	m	bank-full depth
xs_w_0p5_m	m	channel width below bed + 0.5 m
xs_w_1p0_m	m	channel width below bed + 1.0 m

Field	Unit	Meaning
xs_w_2p0_m	m	channel width below bed + 2.0 m
xs_side_slope_l	H:V	left side slope looking downstream
xs_side_slope_r	H:V	right side slope looking downstream
xs_station_elev_json	m	transect as [[station, z], ...] (JSON)
xs_quality	-	high / medium / low
xs_note	-	why the quality was lowered; widths that are lower bounds
tc_kirpich_min	min	Tc by Kirpich (1940)
tc_kerby_kirpich_min	min	Tc by Kerby (1959) overland + Kirpich channel
tc_scs_lag_min	min	Tc by the SCS / NRCS lag equation (NEH 630 ch. 15)
tc_tr55_min	min	Tc by TR-55 segments (sheet + shallow + channel)
tc_bransby_williams_min	min	Tc by Bransby-Williams
tc_kirpich_flag	-	within / outside the Kirpich calibration range
tc_kerby_kirpich_flag	-	within / outside the Kerby range
tc_scs_lag_flag	-	within / outside the SCS lag range
tc_tr55_flag	-	within / outside (channel section quality low)
tc_bransby_williams_flag	-	'rural catchments' (no hard limit)
tc_basis_json	-	inputs used by each method (JSON)
tc_note	-	why a method is empty (missing input)
map_river_dist_m	m	distance from the outlet to the nearest mapped waterway
map_river_name	-	name of that waterway (name field, if given)
map_agrees	0/1	1 when the outlet is within the tolerance of a mapped waterway
map_note	-	why map_agrees is 0 or NULL
unc_area_p10	km2	p10 of the catchment area over the DEM-error realisations
unc_area_p50	km2	p50 of the catchment area over the DEM-error realisations
unc_area_p90	km2	p90 of the catchment area over the DEM-error realisations
unc_area_cv	-	coefficient of variation of the catchment area over the DEM-error realisations
unc_lfp_p10	m	p10 of the longest flow path length over the DEM-error realisations
unc_lfp_p50	m	p50 of the longest flow path length over the DEM-error realisations
unc_lfp_p90	m	p90 of the longest flow path length over the DEM-error realisations
unc_lfp_cv	-	coefficient of variation of the longest flow path length over the DEM-error realisations
unc_s1085_p10	m/m	p10 of the 10-85 slope over the DEM-error realisations
unc_s1085_p50	m/m	p50 of the 10-85 slope over the DEM-error realisations
unc_s1085_p90	m/m	p90 of the 10-85 slope over the DEM-error realisations
unc_s1085_cv	-	coefficient of variation of the 10-85 slope over the DEM-error realisations
unc_tc_p10	min	p10 of the Kirpich Tc over the DEM-error realisations
unc_tc_p50	min	p50 of the Kirpich Tc over the DEM-error realisations
unc_tc_p90	min	p90 of the Kirpich Tc over the DEM-error realisations
unc_tc_cv	-	coefficient of variation of the Kirpich Tc over the DEM-error realisations

Field	Unit	Meaning
unc_lost_pct	%	share of realisations where the outlet lost its stream
unc_switch_pct	%	share of realisations where the area moved by more than 25 %
unc_n	-	number of realisations
unc_note	-	lost outlet or switching, in words

catchments (126 fields)

Field	Unit	Meaning
outlet_uid	-	link key (see crossings)
catchment_id	-	catchment identifier (default = outlet_uid)
outlet_id	-	pour-point feature id - NOT stable
catchment_mode	-	'full' = entire upstream area (overlapping); 'local' = non-overlapping local area
area_km2	km2	contributing area
cells	-	number of DEM cells in the catchment
elev_max_m	m	highest elevation
elev_min_m	m	lowest elevation
elev_mean_m	m	mean elevation
relief_m	m	elev_max_m - elev_min_m
catch_slope_horn	m/m	mean terrain slope over the catchment
catch_relief_ratio	m/m	relief ratio (basin-steepness index)
lfp_length_km	km	longest flow path length (planimetric)
slope_mean	m/m	LEGACY alias of catch_slope_horn (removed in a later release)
slope_relief_ratio	m/m	LEGACY alias of catch_relief_ratio (removed in a later release)
perimeter_km	km	catchment outline length
form_factor	-	form factor A / L^2
elongation_ratio	-	elongation ratio (2/L) sqrt(A/pi)
circularity_ratio	-	circularity ratio 4 pi A / P^2
drainage_density	km/km2	channel length / area
stream_frequency	1/km2	channel links / area
max_strahler	-	highest Strahler order in the catchment
link_method	-	how this feature was linked to its crossing
link_confidence	0-1	confidence of the link
link_note	-	free-text note on the link
status	-	existing = in the design input; proposed = found by the coverage check and delineated in this run, not yet adopted
proposed_reason	-	uncovered_stream / sag_point (proposed crossings only)
nearest_uid	-	nearest existing crossing (proposed crossings only)
nearest_m	m	chainage distance to nearest_uid

Field	Unit	Meaning
soil_sand_pct	%	topsoil sand
soil_silt_pct	%	topsoil silt
soil_clay_pct	%	topsoil clay
soil_oc_pct	%	topsoil organic carbon
soil_cfrag_pct	vol %	coarse fragments
soil_bulk_gcm3	g/cm3	bulk density
soil_texture	-	USDA texture class of the weighted texture
soil_drain_class	-	FAO drainage class with the largest share (E, S, W, M, I, P, V)
soil_hsg_proxy	-	hydrologic soil group PROXY (A-D) from texture and drainage - not a measured infiltration class
usle_k	t.ha.h/(ha.MJ.mm)	USLE/RUSLE K (SI)
usle_k_dg	t.ha.h/(ha.MJ.mm)	alternative K from geometric mean particle diameter
usle_cfrg	-	coarse-fragment factor
soil_coverage_pct	%	share of the catchment covered by soil units with data
soil_dominant_unit	-	soil unit covering the largest share
soil_units	-	number of soil units with data in the catchment
soil_ttr_main	-	taxotransfer rule class of the dominant unit's main profile (SOTWIS TTRmain; confidence indicator)
soil_dataset	-	soil dataset and version
soil_depth_cm	cm	depth interval the soil values describe (default 0-20, D6)
soil_hsg	-	hydrologic soil group with the largest share (A-D)
hsg_pct_a	%	share of the catchment in HSG A
hsg_pct_b	%	share in HSG B
hsg_pct_c	%	share in HSG C
hsg_pct_d	%	share in HSG D, including dual groups (A/D, B/D, C/D)
hsg_pct_dual	%	share in dual groups (high runoff unless drained)
soil_hsg_source	-	source of the HSG values
usle_k_source	-	source of usle_k
cn_ii	-	area-weighted curve number, AMC II
cn_amc	-	antecedent moisture condition of cn_export (I, II, III)
cn_export	-	curve number at cn_amc
cn_coverage_pct	%	share of the catchment with a CN (known cover and HSG)
rational_c	-	area-weighted Rational runoff coefficient
rc_coverage_pct	%	share of the catchment with a Rational C
lc_pct_tree	%	tree cover share
lc_pct_shrub	%	shrubland share
lc_pct_grass	%	grassland share
lc_pct_crop	%	cropland share
lc_pct_built	%	built-up share

Field	Unit	Meaning
lc_pct_bare	%	bare / sparse vegetation share
lc_pct_water	%	permanent water share
lc_pct_wetland	%	herbaceous wetland and mangrove share
lc_pct_other	%	snow / ice and moss / lichen share
cn_lookup_id	-	CN lookup name and version
lc_dataset	-	land-cover dataset
rain_zone	-	rainfall zone with the largest share of the catchment
rain_zone_pct	%	share of the catchment in rain_zone
rain_zones_json	-	share of every zone, % (JSON)
rain_zone_coverage_pct	%	share of the catchment covered by a zone
map_mm	mm/yr	mean annual rainfall over the catchment
map_coverage_pct	%	share of the catchment with rainfall data
map_dataset	-	rainfall dataset
rusle_r	MJ mm/(ha h yr)	RUSLE R ESTIMATED from mean annual rainfall
rusle_r_method	-	R-P relation used for rusle_r
ero_mode	-	RUSLE, or LS-only when R, K or C was not supplied
ero_a_mean_tha	t/ha/yr	mean RUSLE soil loss over the catchment
ero_a_p90_tha	t/ha/yr	90th percentile of RUSLE soil loss
ero_ls_mean	-	mean LS factor
ero_ls_p90	-	90th percentile LS
ero_lnspi_ch_p90	ln(m)	90th percentile ln(SPI) on channel cells
ero_k_mean	t ha h/(ha MJ mm)	area-weighted K
ero_c_mean	-	area-weighted C
ero_p_mean	-	area-weighted P
ero_pct_s1	%	share of cells with combined severity score 1 (lowest)
ero_pct_s2	%	share with combined score 2
ero_pct_s3	%	share with combined score 3
ero_pct_s4	%	share with combined score 4
ero_pct_s5	%	share with combined score 5 (most severe)
ero_rusle_class	-	RUSLE class of the mean soil loss
ero_rusle_score	1-5	severity score of ero_rusle_class
ero_gross_t_yr	t/yr	gross soil loss
ero_sdr	-	sediment delivery ratio
ero_sy_t_yr	t/yr	sediment yield to the crossing
ero_sy_m3_yr	m3/yr	sediment volume to the crossing
ero_bulk_kgm3	kg/m3	bulk density used for the volume
ero_sy_class	-	sediment volume impact class

Field	Unit	Meaning
ero_sy_score	1-5	severity score of ero_sy_class
map_precision	0-1	share of the DEM stream length in the catchment within the tolerance of a mapped waterway
map_recall	0-1	share of the mapped waterway length in the catchment within the tolerance of a DEM stream
map_f1	0-1	harmonic mean of map_precision and map_recall
clip_edge	0/1	1 when the catchment touches the edge of the automatic clip (grid edge or NoData outside the clip mask): enlarge the margin and run again
scenario_name	-	name of the land-cover scenario
cn_ii_scn	-	area-weighted curve number (AMC II) under the scenario
cn_export_scn	-	scenario curve number at cn_amc
d_cn	-	change in curve number
rational_c_scn	-	Rational C under the scenario (user lookup only)
d_rational_c	-	change in Rational C
lc_pct_tree_scn	%	tree share under the scenario
lc_pct_shrub_scn	%	shrub share under the scenario
lc_pct_grass_scn	%	grass share under the scenario
lc_pct_crop_scn	%	crop share under the scenario
lc_pct_built_scn	%	built share under the scenario
lc_pct_bare_scn	%	bare share under the scenario
lc_pct_water_scn	%	water share under the scenario
lc_pct_wetland_scn	%	wetland share under the scenario
lc_pct_other_scn	%	other share under the scenario
ero_c_mean_scn	-	mean C under the scenario
ero_a_mean_tha_scn	t/ha/yr	mean RUSLE soil loss under the scenario
d_ero_a_mean_tha	t/ha/yr	change in mean soil loss

`flowpaths` (34 fields)

Field	Unit	Meaning
outlet_uid	-	link key (see crossings)
flowpath_id	-	flow path identifier (default = outlet_uid)
outlet_id	-	pour-point feature id - NOT stable
lfp_length_km	km	longest flow path length
lfp_length_m	m	longest flow path length
lfp_elev_max_m	m	highest elevation on the path
lfp_elev_min_m	m	lowest elevation on the path
lfp_drop_m	m	lfp_elev_max_m - lfp_elev_min_m
lfp_slope	m/m	whole-path slope

Field	Unit	Meaning
lfp_slope_1085	m/m	10-85 slope along the LFP
lfp_L10_m	m	distance of the 10% point from the outlet
lfp_L85_m	m	distance of the 85% point from the outlet
lfp_z10_m	m	elevation at the 10% point
lfp_z85_m	m	elevation at the 85% point
lfp_z_head_m	m	raw-DEM elevation of the first (headwater) path cell
lfp_z_outlet_m	m	raw-DEM elevation of the last (outlet) path cell
lfp_nonmonotonic	0/1	1 when lfp_drop_m exceeds lfp_z_head_m - lfp_z_outlet_m by more than 0.5 m (spike or pit on the path; lfp_slope slightly overstated)
area_km2	km2	area of the catchment this path belongs to
lfp_overland_m	m	divide → channel head length
lfp_overland_slope	m/m	slope of the overland part
lfp_channel_m	m	channel head → outlet length
lfp_channel_slope	m/m	slope of the channel part
lfp_channel_slope_1085	m/m	10-85 slope of the channel part
lfp_sheet_m	m	sheet-flow part of the overland length
lfp_shallow_m	m	shallow concentrated part of the overland length
lfp_threshold_km2	km2	channel threshold used for the split
lfp_no_channel	0/1	1 when no path cell reaches the threshold (whole path overland)
link_method	-	how this feature was linked to its crossing
link_confidence	0-1	confidence of the link
link_note	-	free-text note on the link
status	-	existing = in the design input; proposed = found by the coverage check and delineated in this run, not yet adopted
proposed_reason	-	uncovered_stream / sag_point (proposed crossings only)
nearest_uid	-	nearest existing crossing (proposed crossings only)
nearest_m	m	chainage distance to nearest_uid

`alignment_profile` (12 fields)

Field	Unit	Meaning
align_name	-	alignment identifier
chainage_m	m	station chainage
x	m	station easting (package CRS)
y	m	station northing (package CRS)
z_dem_m	m	raw DEM ground at the station
z_fill_m	m	filled DEM at the station (empty without a filled DEM)
pond_depth_m	m	depth the fill raised the ground (ponding)
acc_km2	km2	contributing area at the station

Field	Unit	Meaning
stream	0/1	1 at the station nearest a D8 stream crossing
strahler	-	Strahler order of that stream (if an order raster was given)
slope_long_pct	%	longitudinal ground slope, positive = rising chainage
alignment_part	-	part of a multi-part alignment

corridor_sti (12 fields)

Field	Unit	Meaning
ch_start	m	reach start chainage
ch_end	m	reach end chainage
length_m	m	reach length
n_stations	-	stations in the reach
sti_p90_lhs	-	90th percentile of the station sti_p90 on the left side (looking up-chainage)
sti_p90_rhs	-	as sti_p90_lhs, right side
sti_max_lhs	-	highest station sti_p90 on the left side
sti_max_rhs	-	highest station sti_p90 on the right side
sti_class_lhs	-	relative transport capacity of the reach, left side (low / moderate / high / very high; not a severity)
sti_class_rhs	-	as sti_class_lhs, right side
siltation_len_lhs_m	m	length of the reach flagged for siltation, left side
siltation_len_rhs_m	m	as siltation_len_lhs_m, right side

coverage_check (8 fields)

Field	Unit	Meaning
issue	-	missing_crossing / sag_point / flat_stretch / small_area
chainage_m	m	location (start of a stretch)
chainage_to_m	m	end of a flat stretch
area_km2	km2	contributing area (sag: local area against the embankment)
uid	-	outlet_uid of the proposed (or flagged) crossing
nearest_uid	-	nearest existing crossing
nearest_m	m	chainage distance to it
note	-	plain-language finding

drainage_divergence (6 fields)

Field	Unit	Meaning
length_m	m	length of the divergent run
cells	-	stream cells in the run

Field	Unit	Meaning
area_km2	km2	area flowing down the run
rejoins	0/1	1 when the run returns to the mapped course
from_x	m	first cell of the run (where the DEM leaves the map)
from_y	m	

flat_stretches (7 fields)

Field	Unit	Meaning
chainage_m	m	start
chainage_to_m	m	end
length_m	m	length
slope_long_pct	%	mean |longitudinal slope|
crossfall_pct	%	mean cross-fall (steeper side) over crossfall_m
n_crossings_within	-	existing crossings inside the stretch
uids_within	-	their outlet_uids

mapped_rivers_used (2 fields)

Field	Unit	Meaning
name	-	waterway name (name field, if given)
length_m	m	length inside the DEM extent

qeht_run_metadata (52 fields)

Field	Unit	Meaning
schema_version	-	exchange schema (qeht-heas-<major>)
qeht_version	-	QEHT plugin version that wrote the file
run_utc	-	UTC time of the run (ISO 8601)
crs_epsg	-	EPSG code of the (projected, metric) CRS of every layer
crs_name	-	CRS description
cell_size_m	-	DEM cell size (x, y)
dem_path	-	flow-direction / conditioned DEM used for routing
raw_dem_path	-	DEM used for reported elevations and slopes
dem_sha256	-	SHA-256 of the raw DEM file (or size+mtime above the hash limit)
dem_source	-	user description of the DEM (e.g. ALOS 30 m, LiDAR)
dem_prep	-	preparation record of the raw DEM (QEHT_DEM_PREP tag from 'Prepare DEM for hydrology'; empty when not prepared there)
conditioning	-	how the DEM was conditioned (user-declared)
flat_method	-	flat resolution used for flow direction (user-declared)

Field	Unit	Meaning
tie_rule	-	D8 tie-break rule
stream_threshold_cells	-	snap-to-stream threshold in cells
stream_threshold_km2	-	snap-to-stream threshold as area
snap_radius_cells	-	pour-point snap radius in cells
snap_strategy	-	nearest_stream | max_accumulation | none
catchment_mode	-	full (overlapping) | local (non-overlapping)
id_scheme	-	sequential | attribute
id_prefix	-	prefix used for outlet_uid
id_order	-	numbering order for sequential ids
id_attribute	-	pour-point attribute used for ids (attribute scheme)
lfp_1085_reference	-	outlet (v0.9+) | divide (<= 0.8.3)
crossing_source	-	pour points | crossing candidates | relinked package
chainage_start_m	-	start chainage of the road alignment, if one was used
relinked_from	-	package this one was renumbered/relinked from (D3)
soil_dataset	-	soil dataset used for the soil block (empty = none)
soil_depth_cm	-	topsoil depth interval of the soil block
soil_hsg_source	-	source of the hydrologic soil groups (direct dataset or texture proxy)
runoff_json	-	curve number / Rational C inputs: land-cover dataset, condition, AMC, proxy flag and the full lookups (empty = no runoff block)
soil_k_source	-	source of K in the soil block (direct K or Williams/EPIC)
rainfall_json	-	rainfall inputs: zone layer and field, zones, tie rule, QA threshold, rainfall dataset, R-P relation and estimate flag (empty = no rainfall block)
conditioning_burn	-	summary of the breach log when one was supplied (layer burn_log)
erosion_json	-	erosion inputs: mode, factor provenance (source, proxy flags), LS method, class schemes, MCDMA weights, SDR model (empty = no erosion block)
sheet_cap_m	-	sheet-flow cap used for lfp_sheet_m (m)
channel_slope_m	-	approach / exit channel length for ch_slope_us / ch_slope_ds (m)
alignment_source	-	road alignment layer used for chainage and the ground profile
alignment_step_m	-	station spacing of alignment_profile (m)
fp_params_json	-	floodplain width indicator settings: area limit, dz values, bed window, HAND on/off (empty = no road)
quicklook_params_json	-	raster quicklook settings: folder, maximum size (empty = none)
uncertainty_json	-	DEM-error sensitivity: N, sigma and its source (preset), correlation length, seed, threshold, timing, correlation-length sensitivity (empty = not run)
scenario_json	-	land-cover scenario: name, dataset, lookups, erosion relation (empty = none)
autoclip_json	-	automatic corridor clip of the raw DEM (QEHT_AUTOCLIP tag): window, coarse factor, road buffer, margin, mode, memory estimate (empty = not clipped)
mapped_drainage_json	-	check against mapped drainage: source, name field, comparison threshold, tolerance, overall precision / recall / F1 (empty = not run)
tc_params_json	-	time of concentration settings: validity ranges and sources, P2 and its source, channel n, sheet n and Kerby N lookups (PROXY flags), CN used

Field	Unit	Meaning
corridor_sti_params_json	-	side-drain siltation indicator settings: station step, half width, offset step, road strip, drain slope limit, class breaks, slope source (empty = not run)
xs_params_json	-	channel cross-section settings: distance downstream, half width, station step, bank slope, stages (empty = not run)
coverage_params_json	-	coverage check, sag and flat-stretch settings (empty = not run)
n_proposed	-	number of proposed crossings added by the coverage check
n_crossings	-	number of crossings written
parameters_json	-	full parameter dictionary of the run

rasters (14 fields)

Field	Unit	Meaning
name	-	raster id
title	-	display title
png	-	quicklook PNG, relative to the package
world_file	-	world file (.pgw) of the PNG, relative to the package
legend_json	-	legend: classes and colours, or ramp and stretch
crs_epsg	-	EPSG code of the PNG georeference
xmin	m	extent
ymin	m	extent
xmax	m	extent
ymax	m	extent
px_size_m	m	PNG pixel size
source_tool	-	tool that wrote the full-resolution raster
source_raster	-	full-resolution GeoTIFF
kind	-	classified or continuous

sag_points (9 fields)

Field	Unit	Meaning
chainage_m	m	chainage of the low point
z_dem_m	m	ground level
sag_depth_m	m	depth of the sag (prominence of the smoothed minimum)
sag_area_km2	km2	local area draining to the sag (larger side)
sag_area_left_km2	km2	area on the left of the alignment
sag_area_right_km2	km2	area on the right
side	-	contributing side (left / right of increasing chainage)
pond_depth_m	m	fill depth at the station
uid	-	proposed crossing created for the sag (if any)

`xs_transects` (4 fields)

Field	Unit	Meaning
`outlet_uid`	-	link key (see crossings)
`xs_dist_m`	m	distance downstream of the outlet to the section
`xs_station_elev_json`	m	transect as [[station, z], ...] (JSON); station positive on the right looking downstream
`xs_quality`	-	high / medium / low (see crossings)

APPENDIX C

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